The Equilibrium Effects of Automatic Enrollment in Health Insurance*

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Abstract

This paper shows that automatic enrollment rules shape insurer competition and long-run equilibrium outcomes in health insurance markets. We evaluate several enrollment rules following the termination of Colombia's largest insurer, using an equilibrium model in which insurers compete on provider networks and consumers exhibit inertia and adverse selection. Reassigning consumers randomly to incumbent insurers in equal proportions raises social surplus and broadens both hospital and non-hospital networks while modestly increasing spending, because it induces healthy consumers to switch actively in the year after reassignment. Failing to endogenize insurers' supply decisions reverses the conclusions about which rules improve welfare.

Keywords: Automatic enrollment, Health insurer competition, Adverse selection, Inertia.

JEL codes: I10, I11, I13, I18.

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1 Introduction

A common policy tool in health insurance markets is automatic enrollment, whereby individuals are assigned to a health plan when they first become eligible for coverage or when their incumbent plan closes, but are allowed to switch typically after a year (Dorn et al., 2018).[1] This policy tool has recently received attention in the academic literature for its potential to impact consumer welfare and market outcomes (e.g., McIntyre et al., 2021; Agafiev Macambira et al., 2026; Wallace, 2023; Shepard and Wagner, 2023). However, most of the empirical evidence studies automatic enrollment as a static phenomenon without considering its long-run consequences or how insurers' supply decisions might respond.

In this paper, we show that automatic enrollment rules shape competition in health insurance markets and their long-run outcomes, and that failing to endogenize insurers' supply decisions could lead to fundamentally different conclusions about which rules improve welfare. To see why, note that if consumers were reassigned permanently, an enrollment rule would simply determine each insurer's stock of enrollees and its welfare effects would reduce to a matching problem. But automatic enrollment is temporary: consumer switching after the reassignment period shapes how insurers respond to the rule, so welfare depends not only on the quality of the initial match but on the competition the rule sets in motion.

The mechanisms behind the long-run impacts of automatic enrollment are the well-studied phenomena of inertia and adverse selection (Einav and Finkelstein, 2011; Abaluck and Gruber, 2011; Handel, 2013; Ericson, 2014; Polyakova, 2016). When inertia disproportionately affects sick consumers, enrollment rules that induce healthy consumers to switch after reassignment intensifies insurer competition for these profitable switchers, and equilibrium plan characteristics become shaped by the preferences of the healthy (much in the spirit of Ho et al., 2017). This raises consumer welfare when those preferences favor plan characteristics that benefit all consumers, such as broad healthcare provider networks.[2]

[1] Automatic enrollment is used in Medicaid managed care (in states like CA and NY) when individuals who become eligible for Medicaid fail to engage in active plan choice (Agafiev Macambira et al., 2026), Medicare Advantage after an incumbent plan closes Abaluck et al. (2021), and in the US health insurance exchanges to renew insurance coverage through a particular health plan (Anderson et al., 2021).

[2] Conversely, rules that induce sick consumers to switch tend to reduce welfare. In that case, since

Our empirical setting is the Colombian healthcare system, where a health insurer with strong geographical presence, called SaludCoop, was abruptly terminated by the government in December 2015 (Superintendencia Nacional de Salud, 2018). At the time of its termination, SaludCoop operated in 43% of municipalities nationwide and covered nearly 50% of enrollees in some of these markets. The termination was sudden, politically motivated, and unrelated to SaludCoop's overall performance.[3] The government assigned SaludCoop enrollees to a smaller insurer, Cafesalud, during the first three months of 2016, after which enrollees could switch (Superintendencia Nacional de Salud, 2015).

This setting provides a unique opportunity to study automatic enrollment rules, but it also presents some challenges. The sheer magnitude of SaludCoop's termination allows us to estimate effects for the majority of the country and to identify parameters that we would otherwise not have been able to identify (e.g., separating state dependence from preference heterogeneity in consumer inertia). However, we do not have empirical variation in which insurers individuals are assigned to or in enrollment rules themselves (as in other settings, e.g., Agafiev Macambira et al., 2026; Wallace, 2023; Geruso et al., 2023), since all SaludCoop enrollees were assigned to a single incumbent insurer. Our methodological contribution therefore lies in tracing the effects of counterfactual enrollment rules using a structural model of insurer competition.

Our data encompass individual-level enrollment and health claims records from the Colombian contributory health system between 2013 and 2017, representing more than 24 million individuals yearly who pay payroll taxes and their dependents.[4] In this system, insurers compete primarily on their networks of providers and negotiated prices to deliver a single health insurance plan. Other aspects of the plan (premiums, cost-sharing, and benefits) are tightly regulated and uniform across insurers, and insurers receive risk-adjusted per capita payments from the government.

We characterize insurers along two dimensions of network breadth: hospitals and non-

insurers cannot attract healthy types, they resort to "harvesting" their stock of enrollees, for example by narrowing their networks.

[3]See https://www.portafolio.co/negocios/empresas/cayo-saludcoop-gigante-sistema-salud-143450

[4]The contributory system encompasses the portion of the population that pays payroll taxes (and their families), while the other half is covered by a subsidized system fully funded by the government.

hospital providers (clinics, physician practices, pharmacies, and diagnostic providers). Since the impact of enrollment rules depends on how inertia and adverse selection vary across consumers, we begin by documenting this variation. First, we show that insurer switching rates respond far more to changes in network breadth among SaludCoop enrollees than among the general population, demonstrating the existence of inertia. Second, we show that sicker consumers sort into insurers with broader hospital networks but not into those with broader non-hospital networks, consistent with adverse selection on hospital networks specifically. The presence of inertia and adverse selection suggests that automatic enrollment can affect insurer competition by shaping which consumers actively choose their insurer after reassignment.

We provide causal evidence of the impact of the termination on consumer switching and insurer competition using an event study design. We find that switching rates rose 43% in markets with SaludCoop presence (treated) versus those without (control). Treated markets consequently experienced a 22% decline in insurer concentration. Aside from these broad termination effects, the specific assignment rule to Cafesalud also impacted outcomes. Comparing switching rates among SaludCoop enrollees based on the quality of their match to Cafesalud, we find that low-match quality consumers switched at higher rates after the termination. Enrollment rules therefore shape competition through the quality of the initial matches they create.

To explore counterfactual enrollment rules, we specify an equilibrium model of insurer competition on hospital and non-hospital networks. In the model, insurers simultaneously choose their network breadths to maximize the present discounted value of their profits. The profit function incorporates a random utility model of demand in which consumers choose their insurer based on hospital and non-hospital network breadth, out-of-pocket costs, and inertia, with all parameters allowed to vary by diagnosis. On the supply side, insurers' marginal and fixed costs are non-linear functions of both network types. Insurer profits evolve according to exogenous transition probabilities across diagnoses and endogenous transition probabilities across insurers. Our equilibrium concept is a static Nash equilibrium.

Demand estimation faces two identification challenges. First, network breadth and

healthcare costs, which determine out-of-pocket costs, are jointly determined by insurers in anticipation of consumer demand, generating simultaneity bias. Second, the inertia indicator conflates state dependence with preference heterogeneity, making it difficult to identify the welfare-relevant component of inertia. We address the first challenge using an instrumental variable approach and the second exploiting the exogenous reassignment of SaludCoop enrollees to Cafesalud, whose network decisions were independent of any pre-existing consumer preferences for that insurer.

Demand estimates show that consumers with chronic conditions value hospital networks more, and non-hospital networks less, than healthy consumers. Inertia is substantial as consumers are nearly four times more likely to choose their previous insurer, and this parameter reflects primarily switching costs rather than preference heterogeneity. Notably, consumers with chronic conditions are willing to forgo broader hospital and non-hospital networks in exchange for remaining with their incumbent insurer, more so than consumers without chronic diseases.

We estimate the supply side on pre-termination data to later test the model's out-of-sample fit. We find that insurers have heterogeneous marginal and fixed costs. Marginal costs increase with hospital network breadth but decrease with the interaction between the two network types, reflecting economies of scope. Fixed costs increase with both network types, but the effect is considerably larger for non-hospital networks. Hence, the administrative burden of expanding non-hospital coverage—which requires contracting with a larger number of individual providers—drives fixed cost heterogeneity across insurers. Before using the model for counterfactual simulations, we show that it performs well out-of-sample, that is, it accurately predicts the observed distribution of hospital and non-hospital network breadth in 2016 under the observed enrollment rule that assigned SaludCoop enrollees to Cafesalud.

The counterfactual automatic enrollment rules we examine next include random enrollment, enrollment in the insurer with the broadest network, consumer own-choice without a specified insurer default, among others. To predict each rule's equilibrium effects, we first ask which consumers are most likely to switch in the first year following reassignment. In this partial equilibrium exercise, rules with random assignment induce

more switching among relatively more profitable consumers than rules with deterministic assignment, suggesting the former would expand network breadth in equilibrium relative to the latter.

Simulating the full equilibrium confirms these partial-equilibrium predictions. Random enrollment in equal proportions to all insurers generates the highest social surplus of any rule, 44% above the observed scenario, and does so with the smallest increase in healthcare spending per enrollee. It is also the only rule that largely eliminates narrow hospital networks, shifting the mass of insurer–markets with hospital network breadth below 0.5 to the top of the distribution. Moreover, it moves the entire distribution of non-hospital networks to the right. The mechanism that explains these results is the low match quality between consumers and their assigned insurer: mismatched consumers switch, the healthy disproportionately so because the sick place a higher value on inertia, and insurers broaden their networks to compete for them even after netting out the effects on out-of-pocket costs.

By contrast, the own-choice rule produces the narrowest hospital and non-hospital networks of any rule. Consumers who choose their own insurer select one they match well with, so they rarely switch even when rivals broaden their networks. With little competition for switchers, insurers harvest their enrollees instead, narrowing networks to reduce costs.

Insurers' endogenous response to the enrollment rules is key to our results. When we do not allow insurers to respond endogenously to the rules by changing their network breadths, the ranking of the rules in terms of consumer surplus changes completely, with random enrollment performing worse than other rules and the own-choice rule performing better. The dominance of random enrollment emerges because it incentivizes insurers to respond by increasing network breadth.

Our paper contributes to several literatures. The first studies assignment mechanisms. In health insurance, this work exploits quasi-random assignment to plans when individuals first become eligible or when their incumbent plan closes, to estimate the causal effects of plan characteristics on healthcare spending and health outcomes (Brot-Goldberg et al., 2023; Agafiev Macambira et al., 2026; Geruso et al., 2023; Abaluck et al., 2021). These

papers hold the assignment rule fixed and study its downstream consequences, yet how alternative rules would perform has received less attention. One exception is Wallace (2023), who shows in New York Medicaid managed care that assigning consumers to insurers covering their existing providers improves satisfaction, holding insurers' supply decisions fixed. Outside health insurance, the design of assignment rules has been studied where the supply side is fixed by construction, for instance, Huitfeldt et al. (2026) compare assignment mechanisms for patients on general practitioner waitlists in Norway, and Beshears et al. (2024); Illanes (2016); Carroll et al. (2009); Madrian and Shea (2001) characterize the consequences of defaults and switching costs in pension plan choice. We fill a gap in this literature asking how outcomes would differ under alternative assignment rules once insurers' supply decisions respond.

Doing so requires a model of how insurers set plan characteristics in the presence of inertia and adverse selection, which is the subject of the second literature we contribute to: endogenous plan design. Work in this space has overwhelmingly focused on how inertia (and adverse selection) impacts premiums (Saltzman et al., 2025; Miller et al., 2019; Ho et al., 2017; Fleitas, 2017; Polyakova, 2016; Ericson, 2014). One reason for this focus is that inertia can generate dynamic incentives that are easily testable in prices and connect to a broader empirical and theoretical literature on invest-then-harvest incentives (Arie and E. Grieco, 2014; Dubé et al., 2009; Farrell and Klemperer, 2007; Klemperer, 1995). The impact of inertia on non-price health plan characteristics is comparatively less studied, with theoretical extensions to provider networks in Lee and Fong (2013). We extend this literature by quantifying the impact of inertia on hospital and non-hospital provider networks, showing that the well-studied pricing incentives extend to non-price attributes.

The counterfactual enrollment rules we evaluate affect equilibrium outcomes by essentially nudging certain consumers into active plan choice after the grace period. This connects to a third line of literature on nudges and ordeals in health insurance (McIntyre et al., 2021), some of which has shown that these mechanisms hurt welfare by exacerbating adverse selection (Shepard and Wagner, 2023; Handel, 2013) while others show the opposite (Ericson et al., 2025; Domurat et al., 2021; Goldin et al., 2021; Wright et al., 2017; Kling et al., 2012). We add to this literature by examining the equilibrium consequences

of nudges, which are comparatively less studied.

2 Setting

Our setting is Colombia's contributory healthcare system, part of the universal healthcare system of the country, for which enrollment is compulsory for individuals who pay payroll taxes (and their dependents). The contributory system covers approximately half of the country's population (Giedion and Uribe, 2009). Enrollees have access to a single health plan defined and funded by the government, which is operated by private and public insurers. Insurers negotiate with providers to determine network inclusions and health service prices, but other plan elements are regulated and uniform across insurers, including premiums, cost-sharing rates and benefits.[5] Hence, provider networks are the main dimension in which insurers compete for enrollees. Consumer premiums are zero; instead, insurers receive annual per-capita risk-adjusted transfers from the government that poorly fit realized healthcare costs (Espinosa et al., 2023; Riascos et al., 2014).[6]

Automatic Enrollment Rules. To maintain their license with the Ministry of Health, insurers must satisfy several requirements: holding risk-based capital above government-defined thresholds, avoiding losses for more than two consecutive years, and maintaining high enrollee satisfaction as measured by Ministry surveys. Failure to meet these conditions may result in forced termination (Decree 780 of 2016).

In December 2015, the government terminated SaludCoop due to its engagement in illegal activities and financial malpractice.[7] SaludCoop diverted nearly $250 billion to investments outside the healthcare system and submitted false health claims to the government for reimbursement. At the time of its termination, SaludCoop covered nearly 20% of enrollees nationwide (around 4 million individuals), who were transferred to an

[5]Cost-sharing rates are indexed to the enrollee's monthly income level but are standardized across insurers and providers. However, a consumer's total out-of-pocket cost may vary across insurers because the coinsurance rates multiply the health service prices that insurers negotiate with providers.

[6]Appendix Figure 1 compares these risk-adjusted transfer against observed healthcare costs, showing that younger, relatively healthier enrollees can be substantially more profitable than older, sicker ones.

[7]See El Colombiano.

incumbent insurer, Cafesalud, during the first three months of 2016. After this 90-day grace period, enrollees were free to switch. Prior to the termination, Cafesalud covered less than 5% of enrollees nationwide.[8]

The government selected Cafesalud as the assignment insurer because it was in relatively good financial standing and operated in similar markets as SaludCoop. To strengthen the capacity to absorb the massive transfer of enrollees, the government also awarded Cafesalud a $70 million loan.[9]

Following SaludCoop's termination, the Ministry of Health issued Decree 780 of 2016, which for the first time established formal rules for reassigning enrollees when a health plan is terminated. Under Article 2.1.11.3, enrollees without chronic conditions are split evenly: half are randomly assigned in equal proportions to insurers operating in their municipality of residence, while the other half are assigned in proportion to each insurer's local market share. Enrollees with chronic conditions are reassigned entirely based on local market share. In both cases, reassigned enrollees must remain with their new insurer for 90 days before they are allowed to switch plans. Although Decree 780 did not apply to the SaludCoop cohort, it codified the kind of automatic enrollment rules that are likely to govern future plan terminations in Colombia and elsewhere. This raises a fundamental question of whether such rules benefit consumers and how insurers react.

3 Data

We use administrative data from the contributory healthcare system encompassing individual-level enrollment (*Base de Datos Única De Afiliados*), linked with health claims (*Base de Suficiencia*) and earnings (*Planilla Integrada de Liquidación de Aportes*) from 2013 to 2017 for the subset of individuals aged 19 or older. The enrollment data is a snapshot of enrollment for every June. If we observe an individual enrolled with insurer A in June of year t and then again in June of year $t + 1$, we assume this individual did not switch their insurer

[8] We focus on SaludCoop's termination in our analysis because it was sudden, politically motivated, and unrelated to its overall performance. While other insurers were terminated between 2013 and 2017, these operated in the subsidized scheme and had flailing profits.

[9] See El Tiempo.

during the months in between. We label consumers with no gaps in coverage between their first and last observed years in the enrollment data as the *continuously enrolled*.

The health claims data report date of service, service code, diagnosis code (following the International Classification of Diseases, ICD-10), provider identifier, insurer identifier, and negotiated price of the service. Using the diagnosis codes that accompany each claim and following Riascos et al. (2014)'s categorization, we classify individuals as having one of the following health conditions: Cancer, Cardiovascular disease, Diabetes, Pulmonary disease, Renal disease, other diseases, or no diseases. When an individual has multiple diagnoses, we assign the one that accounts for the largest share of their healthcare cost.

We complement enrollment and claims information with data from insurers' public income statements submitted to the National Health Superintendency. These files report insurers' total administrative costs, which we use in some of our descriptive analysis as measures of fixed costs.

We focus on the 13 largest insurers in the contributory system, which account for 99% of enrollees in 2015.[10] For these insurers, we have data on their network of covered providers between 2013 and 2017, obtained from the National Health Superintendency. The provider network data lists the hospitals, clinics, and physician practices included in each insurer's network, but do not specify the particular services for which each provider is covered. We complement these data with networks inferred from health claims, defining a provider as in-network if it submits more than 100 claims for a given insurer.[11]

Given that networks are insurers' main dimension of competition, we characterize each insurer along two dimensions of network breadth: hospitals and non-hospital providers (clinics, pharmacies, physician practices, and diagnostic providers). Hospital network breadth is defined as the fraction of hospitals in a market covered by the insurer. Non-

[10]Excluding SaludCoop and Cafesalud, we observe all 11 remaining insurers for at least three years, 8 for four or more years, and 7 for the full five years. Insurers are missing from the health claims data for the years in which they fail to pass the Ministry of Health's data quality filters. Although we do not observe Cafesalud in 2017, we are able to follow 78% of SaludCoop's enrollees over the full sample period because they had switched out of Cafesalud by then.

[11]Providers identified in the claims data may be absent from the official listings due to timing differences in data submission: while the provider listings are submitted at the beginning of the year (Resolution 1441 of 2016), insurers may add or drop providers throughout the year. In our final network data, one quarter of insurer-provider pairs are inferred from health claims.

hospital network breadth is defined analogously. We identify hospitals as facilities with an intensive care unit, and define markets as municipalities. Our main analysis focuses on the 13 largest municipalities defined by the National Administrative Department of Statistics (DANE), where we observe meaningful variation in both hospital and non-hospital networks and which represent 41% of the country's population.[12]

Appendix Table 1 presents summary statistics for the continuously enrolled in the 13 largest municipalities. Continuous enrollment ensures annualized costs are free of measurement error from partial-year spells, and large municipalities limit spillovers from consumers who travel across markets to receive care. Just under 60% of the sample is healthy, with cardiovascular disease, cancer, and diabetes the most common conditions; annual healthcare costs average 792 thousand pesos ($257 in 2016), 157 thousand of which are out-of-pocket. The table further compares this sample with all individuals and with all continuously enrolled individuals.

Appendix Figure 2 shows the distribution of hospital and non-hospital network breadth in the 13 largest municipalities. Average hospital network breadth equals 0.47, while average non-hospital network breadth equals 0.20. Non-hospital networks are narrower on average because there are more clinics and physician practices than hospitals in any given market.

4 Reduced-Form Patterns

4.1 Network breadth preferences and costs

Insurers' network responses to automatic enrollment rules depend not just on the initial match but on which consumers each rule induces to switch after reassignment. This section documents the two forces governing these switching decisions – inertia and adverse selection – and shows that the costs of expanding each network type mediate the magnitude of insurers' responses.

Figure 1 examines the sensitivity of enrollment decisions to each type of network,

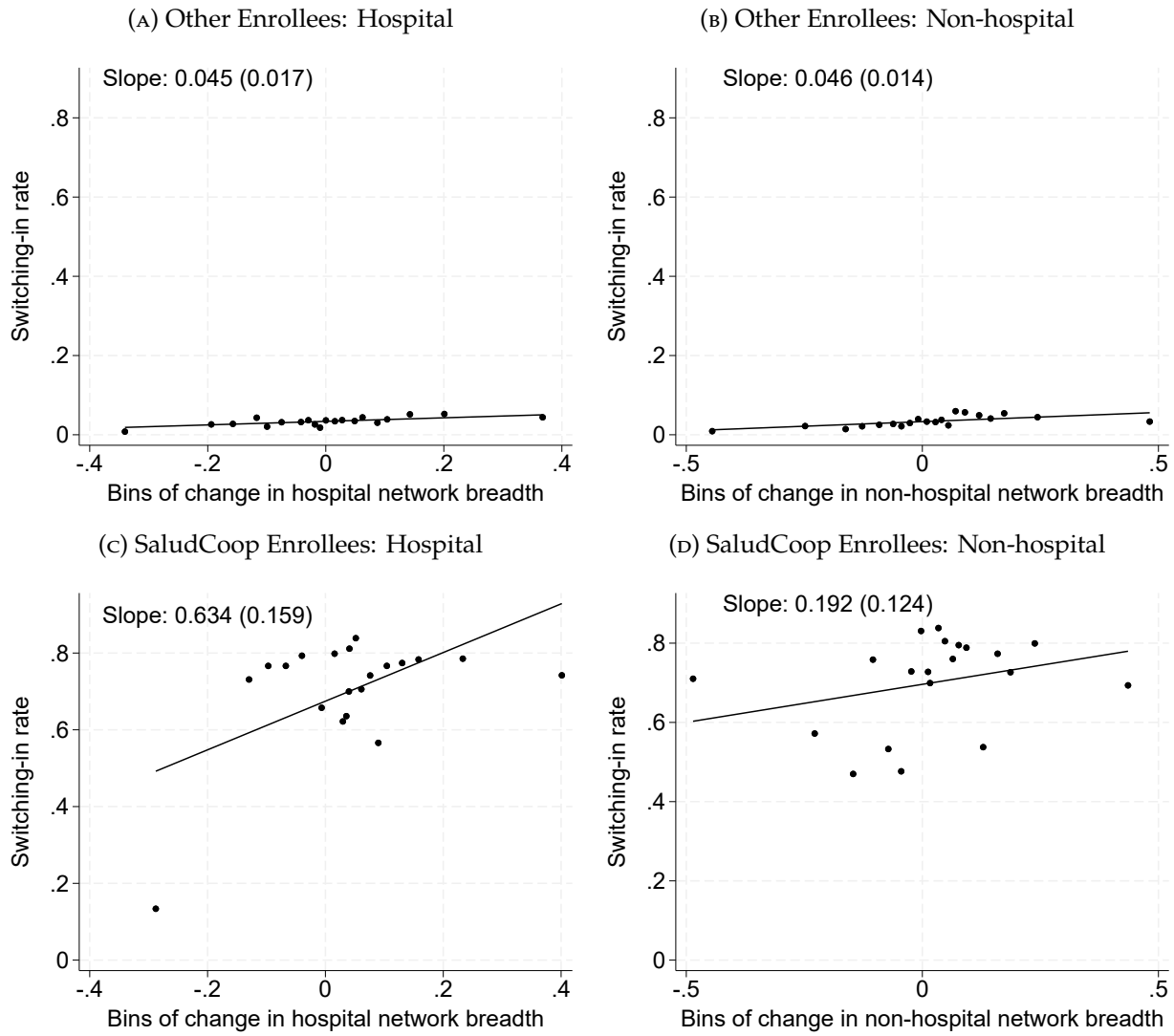
[12]These municipalities are Bogotá, Medellín, Cali, Barranquilla, Bucaramanga, Manizales, Pereira, Cúcuta, Pasto, Ibagué, Montería, Cartagena, and Villavicencio.

separately for non-displaced enrollees (those enrolled to other incumbent insurers) (Panels A and B) and for enrollees displaced from SaludCoop (Panels C and D). Each panel presents a scatterplot of the average switch-in rate by bins of the change in network breadth. The slope coefficient and standard error are reported in the upper left corner of each panel.

The patterns are striking. Among non-displaced enrollees, consumers are more likely to switch into insurers that expand their hospital and non-hospital networks, with slope coefficients of 0.045. Both correlations are precisely estimated but small in magnitude, consistent with inertia that dampens sensitivity to network breadth among the general population.

Among SaludCoop enrollees, the patterns are qualitatively similar but substantially larger. The correlation with hospital network breadth is more than ten times larger than among non-displaced enrollees, and the correlation with non-hospital network breadth is about six times larger. Two features of Panels C and D stand out relative to Panels A and B. First, there is a much higher switching rate among SaludCoop enrollees than non-displaced enrollees, reflecting the absence of persistent preferences for Cafesalud in the former. Because SaludCoop enrollees were forced into Cafesalud for only three months before re-choosing, their subsequent decisions capture network preferences more cleanly than in Panels A and B, where state dependence may dampen sensitivity to network breadth. Second, SaludCoop enrollees respond more strongly to hospital network expansion than to non-hospital network expansion compared to the general population. This is consistent with the hypothesis that SaludCoop enrollees were in worse health than the typical consumer and therefore placed greater weight on access to hospital care.

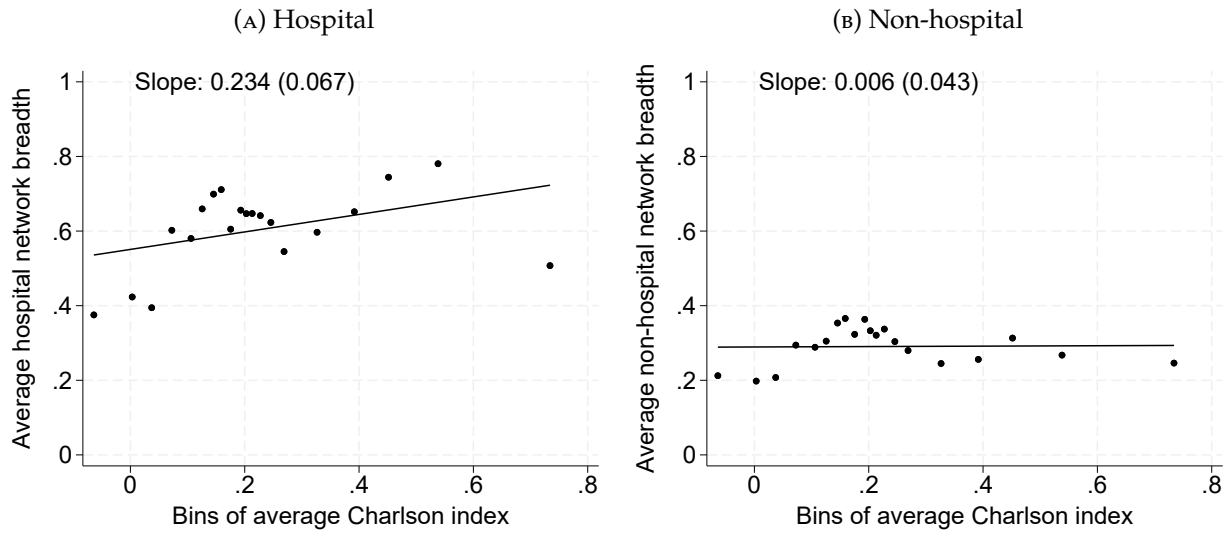
FIGURE 1: Correlation Between Switching Rate and Change in Network Breadth



Note: Figure presents a scatter plot of the average switch-in rate by 20 equally sized bins of the change in hospital network breadth (left column) and non-hospital network breadth (right column) over time. Panels A and B condition on consumers enrolled with incumbent insurers. Panels C and D condition on consumers enrolled with SaludCoop in 2015. To construct this figure, we aggregate our preferred sample to the insurer-market-year level, calculating average switch-in rates and average changes in hospital and provider network breadth. Then, we produce the binned scatterplots after residualizing each variable with market-by-year fixed effects. Solid lines represent linear fits and the numbers in the upper corner of each panel report the slope coefficient and standard error from this regression.

Figure 2 confirms this interpretation. Consumers with a higher Charlson Index – a measure of disease burden – tend to sort into insurers with broader hospital networks, but the correlation between the Charlson Index and non-hospital network breadth is essentially zero. Hence, adverse selection operates mainly through hospital networks.

FIGURE 2: Correlation Between Charlson Index and Network Breadth



Note: Figure presents a scatter plot of the average hospital network breadth in Panel A and average non-hospital network breadth in Panel B by 20 equally sized bins of the average Charlson index. To construct this figure, we aggregate our preferred sample to the insurer-market-year level, calculating the average Charlson index and the average hospital and provider network breadth. Then, we produce the binned scatterplots after residualizing each variable with market-by-year fixed effects. Solid lines represent linear fits, and the numbers in the upper corner of each panel report the slope coefficient and standard error from this regression.

The sorting of sicker consumers into broader hospital networks has direct implications for insurers' costs. Appendix Figure 3, Panels A and B show that average healthcare costs are strongly increasing in hospital network breadth but essentially flat with respect to non-hospital network breadth. The fixed cost patterns in Panels C and D tell a complementary story. Fixed costs per capita are slightly decreasing in hospital network breadth but increasing in non-hospital network breadth. This pattern reflects a fundamental difference in the cost structure of establishing hospital and non-hospital networks. Hospitals are few and large, so adding one to the network represents a substantial expansion in coverage that spreads fixed costs over a larger enrollee base, generating economies of scale. In contrast, non-hospital providers are numerous and small, so expanding non-hospital coverage requires negotiating and maintaining contracts with a large number of individual facilities and physician practices, raising fixed costs at the margin.

4.2 Impacts of the Termination on Market Outcomes

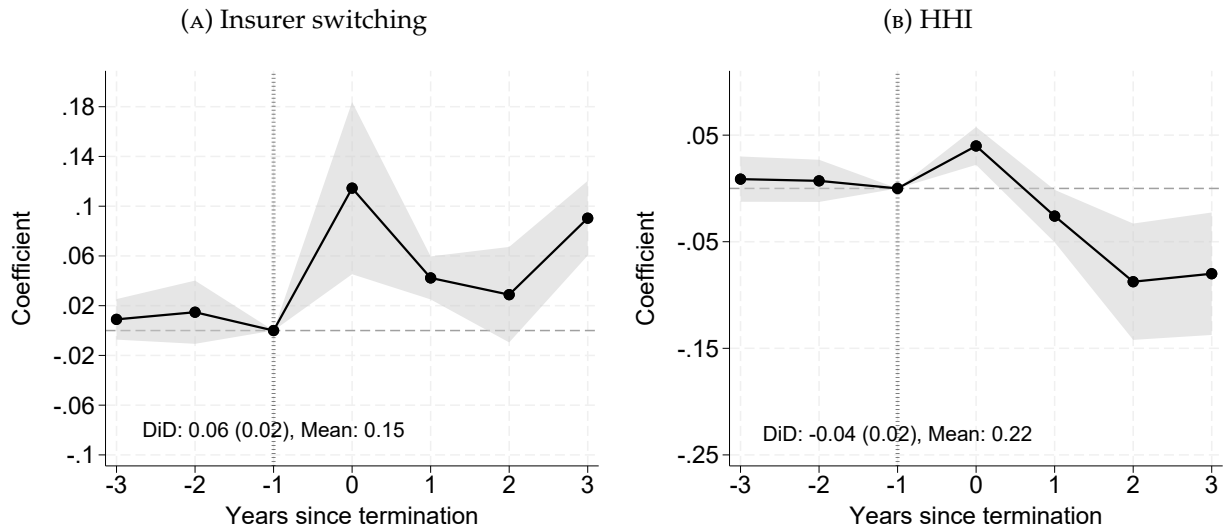
We now use the variation generated by SaludCoop's termination to document how market outcomes changed and whether these changes can be traced to the specific reassignment rule to Cafesalud. We start with the following event study:

$$y_{it} = \beta_{\tau_{m(i)}} + \alpha_i + \delta_{m(i)} + \gamma_t + \varepsilon_{it}$$

where i indexes an individual, t a year, and m a municipality. Treated municipalities are those where SaludCoop operated in 2015; all other municipalities serve as controls. y_{it} is an outcome for individual i in year t, $\tau_{m(i)}$ is the number of years since treatment in i's municipality of residence m (normalized to -1 for control municipalities), and α_i, $\delta_{m(i)}$, γ_t are individual, municipality, and year fixed effects, respectively. For outcomes that vary only at the market level, we estimate the event study on municipality-year-level observations. Standard errors are clustered at the municipality level throughout.

Figure 3, Panel A uses as outcome an indicator for switching insurers. Results show that switching rates in treated municipalities rose by around 10 percentage points in the year of the termination, a 43% increase over the pre-termination mean. The gap relative to control municipalities persisted for three years and widened again in 2019, consistent with enrollees leaving Cafesalud ahead of its own termination that year. Panel B uses as outcome the insurer Herfindahl-Hirschman Index (HHI), measured based on market shares in the number of enrollees. For this analysis, our data is at the municipality-year level. Insurer HHI on average fell 400 points or 22% relative to the pre-termination mean. There is an increase in HHI the year of the termination when Cafesalud received SaludCoop enrollees. Together, the two panels point to a potential mechanism through which automatic enrollment rules can shape market outcomes: rules that induce switching intensify insurer competition.

FIGURE 3: Impact of Termination on Market Outcomes



Note: Figure presents coefficients and 95% confidence intervals of an event study regression where the treatment group are municipalities where SaludCoop operated in 2015. In Panel A an observation is an individual-year and the outcome is an indicator for whether the individual switched insurer. In Panel B an observation is a municipality-year and observations are weighted by the number of enrollees. The dependent variable is the insurer HHI, constructed conditional on insurers that participate in the contributory system and have minimum market share in a municipality greater than 0.002%. The bottom corner of each figure reports the coefficient and standard error of a 2×2 difference-in-difference regression, as well as the mean of the dependent variable among treated municipalities before the termination.

However, the previous design conflates the effects of the termination, the reassignment, and even the loan the government awarded Cafesalud. To provide more direct evidence of the impact of the reassignment rule, separate from aggregate termination effects, we compare switching rates among SaludCoop enrollees who had high vs. low match value with Cafesalud. These match values are the main way in which automatic enrollment rules differ. For example, rules that assign individuals to insurers that cover the providers they had seen will presumably generate higher match values than random enrollment rules.

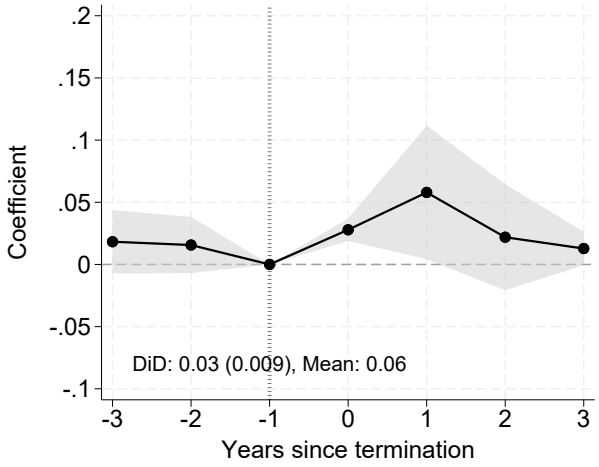
To calculate the match value we estimate the following model:

$$D_{ijt} = \xi_{\theta jm} + \mu_{ijt}$$

where D_{ijt} indicates that individual i was enrolled with insurer j in year t, $\xi_{\theta jm} = \xi_{\text{male},j} + \xi_{\text{age group},j} + \xi_{\text{diagnosis},j} + \xi_{jm}$ collects insurer–sex, insurer–age group, insurer–diagnosis, and insurer–municipality fixed effects, and μ_{ijt} is a type-I extreme value error. The match

value is then $\widehat{\xi}_{\theta jm}$. We split SaludCoop enrollees into those with above- (control) and below-median (treated) match value with Cafesalud in 2015 and estimate an event study comparing the two. We further match treated and control individuals based on pre-period age group, sex, and municipality of residence (exact) and age and income (coarsened). In this case, we cluster standard errors at the individual level.

FIGURE 4: Switching Behavior among SaludCoop Enrollees by Match Value with Cafesalud



Note: Figure presents coefficients and 95% confidence intervals of an event study regression where the treatment group are SaludCoop enrollees with below-median match value with Cafesalud and the control group are SaludCoop enrollees with above-median match value. Match values are measured in 2015. Control individuals are matched to treated individuals based on pre-period age group, sex, and municipality of residence (exact) and age and income (coarsened). The bottom corner of each figure reports the coefficient and standard error of a 2×2 difference-in-difference regression, as well as the mean of the dependent variable among treated individuals before the termination. Standard errors are clustered at the individual level.

Figure 4 presents the results. SaludCoop enrollees with low match values were 3 percentage points more likely to switch than those with high match values, a difference of 50% relative to the pre-termination mean. This suggests that enrollment rules that leave a larger share of enrollees poorly matched will generate more switching. Whether that switching intensifies competition depends on who switches. In Buitrago et al. (2026) we showed that incumbent insurers narrowed their networks after the termination, since SaludCoop enrollees were observably sicker than average and all ended up leaving Cafesalud. Hence, insurers responded to the observed health composition of the pool of active choosers by contracting rather than expanding their networks. The counterfactual rules we evaluate below differ precisely in the composition of switchers they induce.

5 Model

To compare the effects of counterfactual enrollment rules, we develop an equilibrium model of insurer competition. In the model, first, insurers simultaneously choose their hospital and non-hospital network breadths in each market to maximize the present discounted value of profits, taking rivals' choices as given. Then, consumers choose an insurer to enroll with.

5.1 Insurer Demand

The indirect utility of consumer i of type $\theta(i)$ from choosing insurer j in market m and year t is:

$$u_{ijmt} = \sum_k \beta_\theta^k H_{jmt}^k + \alpha_\theta c_{\theta jmt}(\mathbf{H}_{jmt}) + \kappa_\theta \, y_{ijm,t-1} + \xi_{\theta jm} + \nu_{ijm} + \varepsilon_{ijmt} \tag{1}$$

where H_{jmt}^k denotes network breadth for provider type $k \in \{\text{hospital}, \text{non-hospital}\}$. $c_{\theta jmt}(\mathbf{H}_{jmt})$ denotes out-of-pocket costs, which depend on the vector of network breadths $\mathbf{H}_{jmt}$. For each individual, we predict out-of-pocket costs at every insurer in their choice set using the average out-of-pocket costs incurred by patients of the same type θ at insurer j in market m and year t.

$y_{ijm,t-1}$ is an indicator for whether the consumer was enrolled with insurer j in the previous year, and $\kappa_\theta \geq 0$ is a switching cost, incurred whenever the consumer leaves her incumbent. $\xi_{\theta jm}$ is an insurer–consumer type–market fixed effect capturing idiosyncratic preferences of consumer types for insurers as well as unobserved insurer quality that varies across markets; we decompose this term as $\xi_{\theta jm} = \xi_{\text{sex},j} + \xi_{\text{age group},j} + \xi_{\text{diagnosis},j} + \xi_{jm}$.

ν_{ijm} is a persistent consumer–insurer match value, with $\mathbb{E}[\nu_{ijm}|\theta, j, m] = 0$, so that ν_{ijm} is the individual deviation from the type-level taste $\xi_{\theta jm}$. The match value is revealed to the consumer through experience with insurer j but is unobserved to us. Finally, we define consumer type θ as a combination of sex, age group (19–24, 25–29, 30–34, 35–39, 40–44, 45–49, 50–54, 55–59, 60–64, 65–69, 70–74, 75+), income group ($<$ 2, 2–5, $>$ 5 times the monthly minimum wage), and diagnosis (cancer, cardiovascular, diabetes, pulmonary,

renal, other chronic diseases, or no diseases).

Equation (1) contains two sources of inertia. The switching cost κ_θ is a structural friction. The match value ν_{ijm} generates inertia only through sorting: a consumer who has been enrolled with insurer j_0 for some time had compared j_0 to the alternatives and did not leave. Hence, the likelihood of staying with j_0 increases with ν_{ij_0m}. Consumers who remain with an insurer are therefore positively selected on match value:

$$\Delta_\theta \equiv \mathbb{E}\left[\nu_{ij_0m} \,\middle|\, y_{ij_0m,t-1} = 1\right] > 0$$

since each year of continued enrollment screens out low-ν consumers. Δ_θ is not a switching cost but the average taste advantage of the insurers consumers have already sorted into.

Because ν_{ijm} is unobserved, we replace it by its expectation conditional on the consumer's enrollment history. This expectation takes three values:

1. For a non-incumbent alternative, $\mathbb{E}[\nu_{ijm} \mid y_{ijm,t-1} = 0] = 0$.
2. For a consumer choosing an incumbent she selected herself, $\mathbb{E}\left[\nu_{ijm} \,\middle|\, y_{ijm,t-1} = 1\right] = \Delta_\theta$.
3. For a former SaludCoop enrollee choosing Cafesalud (CF) in 2016, it is zero: every SaludCoop enrollee was reassigned to CF, so incumbency carries no information about $\nu_{i,\text{CF},m}$: $\mathbb{E}\left[\nu_{i,\text{CF},m} \,\middle|\, y^{SC}_{i,\text{CF},m,2015} = 1\right] = 0$, where $y^{SC}_{ijm,t-1}$ is the interaction of $y_{ijm,t-1}$ with an indicator for SaludCoop enrollees in 2016. Whatever selection on match value these consumers had accumulated applied toward SaludCoop, which no longer exists, and three months with Cafesalud is too short to have generated a survivor pool.

Since $y^{SC}_{ijm,t-1} = 1$ implies $y_{ijm,t-1} = 1$, the three cases collapse to a single expression,

$$\mathbb{E}\left[\nu_{ijm} \,\middle|\, y_{ijm,t-1}, y^{SC}_{ijm,t-1}\right] = \Delta_\theta\left(y_{ijm,t-1} - y^{SC}_{ijm,t-1}\right) \tag{2}$$

which equals Δ_θ for a self-selected insurer and zero otherwise. Substituting equation (2) into equation (1), the incumbency terms become $\kappa_\theta y_{ijm,t-1} + \Delta_\theta y_{ijm,t-1} - \Delta_\theta y^{SC}_{ijm,t-1}$, and

collecting them yields the utility function we estimate:

$$u_{ijmt} = \underbrace{\sum_k \beta_\theta^k H_{jmt}^k + \alpha_\theta c_{\theta jmt}(\mathbf{H}_{jmt}) + \lambda_\theta y_{ijm,t-1} + \phi y_{ijm,t-1}^{SC} + \xi_{\theta jm}}_{d_{ijmt}} + \varepsilon_{ijmt} \tag{3}$$

with

$$\lambda_\theta = \kappa_\theta + \Delta_\theta, \qquad \phi = -\Delta_\theta \leq 0$$

The coefficient on incumbency for the general population, λ_θ, conflates the switching cost with match-value selection; the coefficient on incumbency for former SaludCoop enrollees in 2016, $\lambda_\theta + \phi$, isolates the switching cost. The two components are separately identified since $\hat{\kappa}_\theta = \hat{\lambda}_\theta + \hat{\phi}$ and $\hat{\Delta}_\theta = -\hat{\phi}$. Equation (3) also restricts Δ_θ to be common across types while allowing κ_θ to vary across types. The deviation $\nu_{ijm} - \mathbb{E}[\nu_{ijm}|\cdot]$ is absorbed into ε_{ijmt}.

This decomposition rests on two restrictions. First, κ_θ does not depend on termination status, that is, the administrative cost of changing insurers is the same whether or not one's previous insurer was terminated. Second, Cafesalud is not a systematically better match for former SaludCoop enrollees than a randomly drawn insurer would be, $\mathbb{E}\left[\nu_{i,\text{CF},m} \,\middle|\, y_{i,\text{CF},m,2015}^{SC} = 1\right] = 0$. One potential violation of this restriction is that Cafesalud was required to maintain SaludCoop's provider network during the reassignment period. However, the requirement lasted only three months, and our utility function already controls for network breadth.

The distinction between κ_θ and Δ_θ matters for the welfare comparisons in Section 7. Only Δ_θ reflects genuine preferences and enters consumer surplus, whereas κ_θ is a friction that we exclude from the calculation, following Handel (2013).

To estimate this demand function, we use consumers' out-of-pocket costs as directly obtained from the data.[13] But in counterfactuals, to appropriately capture the cost-coverage

[13]For consumers who earn less than 2 times the monthly minimum wage (MMW), the coinsurance rate is 11.5% of the service price, the copay is 11.7% times the daily minimum wage, and the out-of-pocket maximum is 57.5% times the MMW. For those with incomes between 2 and 5 times the MMW, the coinsurance rate is 17.3%, the copay is 46.1% times the daily minimum wage, and the out-of-pocket maximum is 230% times the MMW. Finally, for those who earn 5 or more MMWs, the coinsurance rate is 23%, the copay is 121.5% of the daily minimum wage, and the out-of-pocket maximum is 460% times the MMW.

trade-off that consumers face (i.e., broader networks are related to higher out-of-pocket costs as seen in Appendix Figure 4), we allow the out-of-pocket cost to depend on both network types:

$$c_{\theta jmt} \equiv r_\theta C_{\theta jmt}(\mathbf{H}_{jmt})$$

where r_θ is the coinsurance rate and $C_{\theta jmt}(\mathbf{H}_{jmt})$ is the insurer's marginal cost, described in the next subsection.

Assuming the preference shock ε_{ijmt} is distributed T1EV, the consumer's choice probability is given by:

$$s_{ijmt} = \frac{\exp(d_{ijmt})}{\sum_{k \in \mathcal{J}_{mt}} \exp(d_{ikmt})}$$

where $\mathcal{J}_{mt}$ is the set of insurers that operate in market m in year t. Note that there is no outside option as enrollment is mandatory in Colombia.

Identification. An additional source of endogeneity that may bias our demand estimates is the correlation between unobserved health status, plan choice, and out-of-pocket costs. Individuals with worse unobserved health may select into insurers with broader networks and incur higher out-of-pocket costs conditional on enrollment, which may attenuate both the estimated disutility of out-of-pocket costs and the preference for network breadth.

We address the endogeneity of out-of-pocket costs by instrumenting with the fraction of an insurer's non-hospital providers that exit its network because they shut down, interacted with the coinsurance rate. Exits shift bargaining leverage toward the providers that remain, raising the prices insurers negotiate and hence the out-of-pocket costs their enrollees face. Because enrollment is mandatory, exits do not affect total demand, and conditional on network breadth and the rich set of fixed effects, they affect utility only through negotiated prices. The interaction with the coinsurance rate further exploits the exogeneity of the individual's income level.

Finally, network breadth could be endogenous for two reasons. First, insurers with broader networks may have higher unobserved quality, so that H^k_{jmt} is correlated with ξ_{jmt}; insurer–market fixed effects absorb this, and identification uses changes in breadth over time. Second, those changes could themselves be responses to unobserved demand

(simultaneity bias). We address this with two additional instruments: the fraction of an insurer's hospitals that exit its network because they shut down, which shifts breadth for the reasons above, and an indicator for whether the insurer covers a public hospital, which shifts breadth because some insurers are required by law to contract a share of their network with public hospitals (Congreso de la República de Colombia, 2007).[14]

5.2 Supply and Nash Equilibrium

If consumer utility depended only on network breadth and reassignment were permanent, the optimal enrollment rule would be trivial: assign consumers to the insurer with the broadest networks. Two features of our setting complicate this prediction. First, consumers face a trade-off between network breadth and out-of-pocket costs, so broader coverage does not unambiguously raise welfare. Second, insurers respond endogenously to the composition of their assigned enrollees. Therefore, the welfare consequences of any enrollment rule depend on how insurers adjust their networks.

Insurer's problem. Insurers compete in network breadth in each market m. Time is discrete. In each period, insurer j chooses a vector of network breadths $\mathbf{H}_{jm} = (H_{jm}^{\text{hosp}}, H_{jm}^{\text{non-hosp}})$ given rivals' choices $\mathbf{H}_{-j,m}$, with $\mathbf{H}_m = \{\mathbf{H}_{jm}\}_{j=1}^{J_m}$. The state of the market is the distribution $\mathbf{N}$ of consumers over types θ, incumbency status y, and insurers. Insurer j's per-enrollee profit is:

$$\pi_{jm}(\mathbf{H}_m, \theta, y) = \left(R_{\theta m} - (1 - r_\theta) C_{\theta jm}(\mathbf{H}_{jm})\right) s_{jm}(\mathbf{H}_m, \theta, y)$$

where $R_{\theta m}$ is the risk-adjusted transfer from the government (plus average copayments), r_θ the coinsurance rate, $C_{\theta jm}$ the insurer's marginal cost, and s_{jm} the choice probability from the demand model. The risk-adjusted transfer encompasses a compensation for the enrollee's demographic characteristics (sex, age, and municipality of residence), a

[14] Article 16 of Law 1122 of 2007 requires insurers operating in the subsidized regime to contract at least 60% of their risk-adjustment-financed spending with public hospitals. Three insurers in our sample operate in both regimes, and because they maintain these contracts for their subsidized enrollees, their contributory networks are more likely to include public hospitals for reasons unrelated to contributory demand. We use this as the source of variation behind our hospital network instrument.

compensation for a coarse list of diseases (known as the High-Cost Account), and disability compensations.[15] Maintaining a network of type k costs $F^k_{jm}(H^k_{jm})$ per period.

Between periods, an enrollee of type θ remains in the contributory system with probability $1-\rho_\theta$,[16] transitions to type θ', and chooses an insurer for the following year. We denote by $\mathcal{P}(\theta', j \mid \theta, y)$ the probability that a consumer of type θ enrolled with insurer y in period t transitions to type θ' enrolled with insurer j in period $t+1$. Insurer j's value function satisfies:

$$V_j(\mathbf{N}) = \max_{\mathbf{H}_{jm}} \left\{ \sum_{\theta,y} N_{\theta y m}\, \pi_{jm}(\mathbf{H}_m, \theta, y) - \sum_k F^k_{jm}(H^k_{jm}) + \zeta\, V_j\big(\mathbf{N}'(\mathbf{H}_m, \mathbf{N})\big) \right\}$$

where ζ is the discount factor and $\mathbf{N}'$ is next period's state, generated by $\mathcal{P}$, by dropout of the contributory system, and by rivals' equilibrium choices $\mathbf{H}_{-j,m} = \sigma_{-j}(\mathbf{N})$. The first-order condition for network type k is

$$\underbrace{\sum_{\theta,y} N_{\theta y m} \frac{\partial \pi_{jm}}{\partial H^k_{jm}} - \frac{\partial F^k_{jm}}{\partial H^k_{jm}}}_{\text{static}} + \underbrace{\zeta \frac{\partial V_j}{\partial \mathbf{N}'} \cdot \frac{\partial \mathbf{N}'}{\partial H^k_{jm}}}_{\text{dynamic}} = 0$$

The dynamic term captures two incentives absent from a static model. Because network breadth today shifts enrollment and its health composition tomorrow, and because enrollees are partially locked in by inertia, an insurer can attract consumers with a broad network today and serve them with a narrower one tomorrow – the invest-then-harvest incentive of Farrell and Klemperer (2007). Moreover, because rivals' future networks depend on the state, the insurer internalizes how its enrollment gains change their behavior.

This dynamic term has two testable implications. First, if insurers anticipate how their enrollment base affects future profits, network breadth should adjust to year-to-year variations in enrollment even when the rest of the environment is fixed. Second, insurers

[15] For an explanation of how disability transfers are calculated and other considerations of insurer revenues see Resolution 06411 of 2016 of the Ministry of Health accessed through https://www.minsalud.gov.co/Normatividad_Nuevo/Resoluci%C3%B3n%206411%20de%202016.pdf.

[16] The dropout probability depends on whether the individual is unemployed for several months and hence it is largely unrelated to the breadth of insurers' provider networks.

should narrow their networks after gaining enrollees, as the invest-then-harvest incentive predicts.

We test these predictions in Table 1. Columns (1) and (2) establish the relation between the change in network breadth and the lagged change in total enrollment, for hospital and non-hospital networks respectively. We estimate these regressions with data from 2015, before SaludCoop's termination. We find a precise zero relation between the two variables, suggesting that networks do not respond meaningfully to year-to-year variations in enrollment. Columns (3) and (4) estimate the same regressions with data from 2017, after the termination and after the grace period with Cafesalud. We find precise zeros in this case as well, so invest-then-harvest incentives after gaining SaludCoop enrollees are limited in this market.[17]

TABLE 1: Test of Network Breadth Commitment

	Year 2015		Year 2017	
	$\Delta H_{jmt}^{\text{hosp}}$ (1)	$\Delta H_{jmt}^{\text{non-hosp}}$ (2)	$\Delta H_{jmt}^{\text{hosp}}$ (3)	$\Delta H_{jmt}^{\text{non-hosp}}$ (4)
$\%\Delta N_{jm,t-1}$	-0.0023	0.0002	0.0008	-0.0003
	(0.0022)	(0.0026)	(0.0021)	(0.0025)
$H_{jm,t-1}^{\text{hosp}}$	-0.228***		-0.144**	
	(0.049)		(0.051)	
$H_{jm,t-1}^{\text{non-hosp}}$		-0.237***		-0.192***
		(0.023)		(0.025)
Constant	0.026***	0.153***	0.028***	0.103***
	(0.006)	(0.012)	(0.008)	(0.015)
Market FE	Yes	Yes	Yes	Yes
Insurer FE	Yes	Yes	Yes	Yes
N	549	444	2794	1548
R^2	0.550	0.450	0.552	0.451

Note: Table reports OLS regressions using as outcome variables the change in hospital and non-hospital network breadth over time. The main independent variable is the lagged change in enrollment. An observation is an insurer-market-year. Columns (1) and (2) use data from 2015 before SaludCoop's termination to test for persistence of network breadth within insurer-market. Columns (3) and (4) use data from 2017 to test for incentives to harvest at insurers that gained enrollees after SaludCoop's termination. All specifications include insurer and market fixed effects. Standard errors are clustered at the market level. $^{***}p < 0.001$,$^{**}p < 0.05$,$^{*}p < 0.1$.

Based on this evidence, we characterize the stationary equilibrium of the game under the following assumptions.

[17] A placebo regression of the 2015 change in breadth on the 2016 enrollment gain, in Appendix Table 2 is also zero, confirming that insurers who later gained displaced enrollees were not on a different network trajectory beforehand.

Assumption 1 (Stationary network policies)**.** *Conditional on the market environment, equilibrium network breadth does not respond to year-to-year changes in the distribution of enrollees:* $\sigma_j(\mathbf{N}) = \mathbf{H}_{jm}$ *for all* j *and all* $\mathbf{N}$*.*

Assumption 2 (Fixed market size)**.** *The mass of type-*θ *consumers in market* m*,* $N_{\theta m}$*, is fixed over time.*

Assumption 3 (Separable transitions)**.** $\mathcal{P}(\theta', j \mid \theta, y) = P(\theta' \mid \theta)\, P(j \mid \theta, y)$*.*

Assumption 1 formalizes the evidence from Table 1. This assumption does not require networks to be constant over time (they still respond to changes in the environment such as SaludCoop's termination), only that insurers not re-optimize networks in response to year-to-year changes in their enrollment when the rest of the environment is fixed.[18] Assumption 2 abstracts from incentives to invest in network breadth today to attract future enrollees in expanding markets. Assumption 3 states that a consumer's current health status affects which insurer they enroll with tomorrow, but the probability of transitioning across health statuses is independent of the insurer. A formal derivation of this is presented below:

$$\mathcal{P}(\theta', j \mid \theta, y) = P(\theta' \mid j, \theta, y)\, P(j \mid \theta, y) \quad \text{Chain rule}$$

$$\mathcal{P}(\theta', j \mid \theta, y) = P(\theta' \mid \theta)\, P(j \mid \theta, y) \quad \text{Conditional independence of } \theta' \text{ and } j \text{ given } \theta$$

$$\mathcal{P}(\theta', j \mid \theta, y) = P(\theta' \mid \theta)\, s_{jm}(\mathbf{H}_m, \theta, y) \quad \text{Replacing the choice probability among type } \theta$$

Appendix Table 3 presents evidence for the second step in this derivation showing that conditional on not having a diagnosis in year $t-1$, the probability of being diagnosed with a chronic disease in year t is generally unrelated to hospital and non-hospital network breadth.

Estimable profit function. Under Assumptions 1–3, the profit of insurer j is the sum over its current enrollees of the expected discounted profit each generates while she remains

[18] The assumption also makes the problem tractable. A Markov perfect equilibrium would require solving for policies over a state that is a distribution across 504 consumer types, incumbency status, and up to 13 insurers, with adverse selection making the full composition rather than aggregate enrollment relevant for optimal networks.

with j, net of fixed costs:

$$\Pi_{jm}(\mathbf{H}_m) = \sum_{\theta} \underbrace{\pi_{ijm}(\mathbf{H}_m, \theta, y)}_{\text{per-enrollee profit}} N_{\theta ym} + \underbrace{\sum_{t=1}^{T} \zeta^t \sum_{\theta, y} (1-\rho_\theta)\mathcal{P}(\theta', j|\theta, y)\pi_{ijm}(\mathbf{H}_m, \theta', y)N_{\theta ym}}_{FP_{jmt}} - \underbrace{\sum_k F_{jm}^k}_{\text{fixed cost}} \tag{4}$$

where ζ is set to 0.95 in estimation, and T is the simulation horizon, set to 100. The first term is current profit from the enrollment base; FP_{jm} is the discounted value of retaining that base as enrollees age, transition across diagnoses, and either stay or switch. Because Assumption 1 fixes networks on the equilibrium path, whether F_{jm}^k is a one-time or a per-period cost only rescales the fixed-cost term by $\sum_t \zeta^t$, which is absorbed by the intercept of the estimating equation.

Marginal and fixed costs. In the per-enrollee profit function, the insurer's marginal cost is

$$C_{\theta jmt} = \exp\left(\sum_k \tau_1^k H_{jmt}^k + \tau_3 H_{jmt}^{\text{hosp}} H_{jmt}^{\text{prov}} + \gamma_\theta + \eta_j + \delta_m + \epsilon_{\theta jmt}\right) \tag{5}$$

where γ_θ, η_j, and δ_m are consumer type, insurer, and market fixed effects, respectively,[19] and $\epsilon_{\theta jmt}$ is white noise.

The fixed cost for network type k is:

$$F_{jm}^k = \exp\left(\omega H_{jm}^k + \nu_{jm}^k\right)$$

where ν_{jm}^k is the model's structural error. We allow this error to differ across network types because, as seen in Appendix Figure 3 Panels C and D, economies of scale are heterogeneous across hospital and non-hospital networks.

[19] Our functional form for the marginal cost is informed by the descriptive evidence in Figure 3, Panels A and B, that shows that healthcare costs increase at decreasing rates in hospital and non-hospital network breadth.

First-order condition. The first-order condition of the insurer's problem for network type k is:

$$FOC_{jm}^{k}(\mathbf{H}_m) \equiv MP_{jm}^{k}(\mathbf{H}_m) - \omega \cdot \exp(\omega H_{jm}^{k} + \nu_{jm}^{k}) = 0$$

where $MP_{jm}^{k}(\mathbf{H}_m)$ denotes the marginal variable profit associated with network type k, that is, the derivative of the first two terms of equation (4) with respect to H_{jm}^{k}. Taking logs yields:

$$\log(MP_{jm}^{k}(\mathbf{H}_m)) = \log(\omega) + \omega H_{jm}^{k} + \nu_{jm}^{k} \tag{6}$$

which is the regression we take to the data, with $\log(\omega)$ absorbed by the intercept.

Identification. Given the richness and size of our data, we can calculate insurers' marginal costs as the average healthcare cost across all individuals who are of type θ. We then estimate the marginal cost function using a Poisson pseudo-likelihood regression on the sample of all continuously enrolled individuals residing in *any* municipality (Wooldridge, 2023).[20]

The main identification challenge is unobserved patient selection into insurers based on network breadth. For example, unobservably sicker consumers within a given consumer type may choose insurers with broader networks, which would bias our estimates of τ_1^k upward. However, our consumer type definition (with 504 possible combinations) is sufficiently rich to largely address this concern. Within each type, the remaining variation in health status that could drive selection is limited. To verify the direction of the bias, we estimate two versions of the marginal cost function: one that excludes consumer type fixed effects and one that includes them. If unobserved selection is the primary source of bias, the marginal effect of network breadth should be larger in the specification without consumer type fixed effects.

[20]In the estimation of insurer marginal costs we use data from all municipalities rather than the 13 largest municipalities to avoid small cells and potential aggregation bias, especially for consumer types characterized by uncommon chronic diseases such as renal disease.

6 Estimation Results

We now present and discuss our estimation results.

6.1 Insurer Demand

We estimate the demand model on a 10% random sample of continuously enrolled individuals residing in the 13 largest municipalities for dimensionality reasons. Table 2 reports the results. Column (1) omits both the incumbency indicators and the control function, column (2) adds the incumbency indicators, and column (3), our preferred specification, adds the control function. Appendix Table 4 reports the first stages where we see that the instruments are strongly related to the endogenous variables.

The three columns separate the two sources of bias in the network coefficients. Omitting incumbency in column (1) attenuates them toward zero relative to column (2), since inertia makes consumers appear unresponsive to networks and prices. Ignoring simultaneity attenuates them further: the coefficients in column (2) are in turn below those in column (3). Control function coefficients at the bottom of the table are also statistically significant, confirming that hospital network breadth and out-of-pocket costs are endogenous.

Focusing on column (3), we find that consumers on average have strong preferences for broader hospital and non-hospital networks and experience significant disutility from higher out-of-pocket costs. The implied average elasticity of insurer demand with respect to the out-pocket cost is -0.64. Individuals with chronic diseases have much stronger preferences for hospital network breadth but weaker preferences for non-hospital network breadth than their healthier counterparts. Those with chronic disease are also substantially less responsive to out-of-pocket costs and more likely to stay with their incumbent insurer than individuals without diagnoses.

TABLE 2: Insurer Demand Model

	(1) No CF, no incumbency		(2) No CF		(3) Main	
	coef	se	coef	se	coef	se
Hospital network breadth	0.221	(0.078)	0.635	(0.082)	7.942	(0.216)
Non-hospital network breadth	0.150	(0.096)	0.692	(0.099)	0.838	(0.102)
OOP spending	-1.508	(0.106)	-0.966	(0.107)	-5.531	(0.391)
Incumbent			4.212	(0.019)	4.076	(0.023)
Incumbent SC			-0.197	(0.022)	-0.230	(0.037)
Hospital network breadth						
× Male	-0.014	(0.051)	-0.100	(0.052)	-0.089	(0.051)
× Age	0.005	(0.002)	-0.006	(0.002)	-0.006	(0.002)
× Cancer	0.532	(0.102)	0.443	(0.102)	0.465	(0.089)
× Diabetes	0.847	(0.226)	0.998	(0.224)	1.040	(0.241)
× Cardiovascular	1.029	(0.089)	1.083	(0.089)	1.097	(0.087)
× Pulmonary	0.060	(0.332)	0.278	(0.331)	0.316	(0.329)
× Renal	0.713	(0.397)	0.738	(0.393)	0.787	(0.376)
× Other disease	0.759	(0.126)	0.700	(0.125)	0.678	(0.129)
Non-hospital network breadth						
× Male	-0.226	(0.061)	-0.328	(0.063)	-0.335	(0.059)
× Age	0.022	(0.002)	0.008	(0.002)	0.008	(0.002)
× Cancer	-1.302	(0.123)	-1.372	(0.123)	-1.402	(0.111)
× Diabetes	-0.295	(0.270)	-0.271	(0.269)	-0.318	(0.259)
× Cardiovascular	-0.819	(0.108)	-0.834	(0.109)	-0.848	(0.115)
× Pulmonary	0.410	(0.397)	0.392	(0.391)	0.363	(0.417)
× Renal	-0.710	(0.476)	-0.580	(0.473)	-0.609	(0.385)
× Other disease	-0.587	(0.159)	-0.627	(0.158)	-0.631	(0.154)
OOP spending						
× Male	-0.238	(0.044)	-0.230	(0.041)	-0.214	(0.051)
× Age	0.013	(0.001)	0.004	(0.001)	0.004	(0.002)
× Cancer	1.613	(0.106)	1.257	(0.102)	1.169	(0.119)
× Diabetes	1.062	(0.139)	1.016	(0.132)	0.929	(0.147)
× Cardiovascular	1.704	(0.112)	1.318	(0.105)	1.229	(0.117)
× Pulmonary	1.199	(0.111)	1.120	(0.107)	1.033	(0.136)
× Renal	0.971	(0.115)	0.848	(0.107)	0.764	(0.116)
× Other disease	1.384	(0.105)	1.268	(0.101)	1.172	(0.136)
Incumbent						
× Male	0.775	(0.010)	0.115	(0.012)	0.136	(0.012)
× Age	0.140	(0.000)	0.032	(0.000)	0.035	(0.001)
× Cancer	0.634	(0.023)	0.112	(0.024)	0.268	(0.027)
× Diabetes	-0.479	(0.062)	0.424	(0.061)	0.727	(0.055)
× Cardiovascular	-0.201	(0.022)	0.201	(0.022)	0.401	(0.031)
× Pulmonary	-0.116	(0.085)	0.335	(0.084)	0.881	(0.090)
× Renal	0.326	(0.115)	0.598	(0.112)	1.125	(0.124)
× Other disease	0.531	(0.029)	0.391	(0.029)	0.529	(0.032)
CF OOP spending					4.634	(0.383)
CF Hospital network					-7.656	(0.207)
N	13,878,827		13,878,827		13,878,827	
Pseudo-R^2	0.806		0.813		0.813	

Note: Table presents maximum likelihood estimates of the insurer demand model using a conditional logit. Estimation uses a random sample of 500,000 individuals enrolled throughout the sample period from 2013 to 2017 in the 13 largest municipalities: Bogotá, Medellín, Cali, Barranquilla, Bucaramanga, Manizales, Pereira, Cúcuta, Pasto, Ibagué, Montería, Cartagena, and Villavicencio. Column (1) presents the model without control function nor incumbency controls. Column (2) presents the model with incumbency controls but without control function. Column (3) includes both incumbency controls and the instruments for OOP spending and hospital network breadth implemented through a control function. Specifications include insurer, insurer–age group, insurer–diagnosis, insurer–sex, and insurer–market fixed effects. Robust standard errors in parenthesis for columns (1) and (2) and bootstrap standard errors in parenthesis in column (3) based on 50 resamples of individuals.

Patients are nearly four times more likely to choose the insurer they were enrolled with in the previous year. The coefficient on the interaction between incumbency status and SaludCoop enrollees in 2016 is negative, in line with our theoretical framework. This suggests that 94% of the persistence in insurer choice is explained by switching costs, $\hat{\lambda}_\theta + \hat{\phi}_\theta$, and the remaining 6% by preference heterogeneity. Monetizing the overall inertia estimate (using $-\frac{\hat{\lambda}_\theta}{\hat{\alpha}_\theta}$) yields a median switching cost of 1.072 million pesos (1.5 times the monthly minimum wage in 2016 or \$363).

Table 3 summarizes our estimates of consumers' willingness-to-pay (wtp) for a 10 percentage point increase in hospital and non-hospital network breadth (computed as $-\frac{10}{\alpha_\theta}\frac{\partial s_{ijm}}{\partial H^k_{jm}}$ and taking into account the dependence of out-of-pocket costs to network breadth) and for remaining with their incumbent insurer (computed as $-\frac{\hat{\lambda}_\theta}{\hat{\alpha}_\theta}$). Consumers with chronic conditions are willing to pay between 0.4 and 0.7 million pesos (\$135–\$237) more than healthy consumers to remain with their incumbent insurer. As seen in columns (2) and (3), willingness-to-pay is similar across health conditions for hospital networks but much smaller among chronic patients for non-hospital networks. Comparing the estimates across the first three columns shows that individuals value much more staying with their incumbent than having broad networks.

Columns (4) and (5) make this comparison more directly by reporting the ratio of wtp for hospital and non-hospital networks to wtp for remaining with the incumbent insurer. All the estimates are less than one. Among individuals with chronic conditions, the value of remaining with the incumbent insurer is large relative to the value of a significant improvement in hospital and non-hospital network breadth, suggesting that inertia is the dominant force governing their insurer choices. Among individuals without chronic conditions, the relative value of inertia is smaller, indicating that this group is more willing to switch insurers in exchange for broader hospital networks. Appendix Table 5 confirms that the demand model fits the data well in-sample.

TABLE 3: Switching Cost and Willingness-to-Pay for Networks

	WTP remain w/ same insurer (1)	WTP hospital network (2)	WTP non-hosp network (3)	WTP ratio hosp network (4)	WTP ratio non-hosp network (5)
Cancer	1.408	0.309	-0.008	0.219	-0.006
Cardiovascular	1.577	0.240	0.008	0.152	0.005
Diabetes	1.570	0.185	0.019	0.118	0.012
Pulmonary	1.677	0.175	0.036	0.105	0.022
Renal	1.618	0.221	0.026	0.137	0.016
Other disease	1.491	0.239	0.014	0.160	0.009
No disease	1.014	0.281	0.034	0.277	0.033

Note: Table shows the median willingness to pay to remain with the same insurer in column (1), the median willingness-to-pay for a 10 percentage point increase in hospital network breadth in column (2), the median willingness-to-pay for a 10 percentage point increase in non-hospital network breadth in column (3), and the average ratio between columns (2) and (1) and between columns (3) and (1), in columns (4) and (5) respectively. All measures of willingness to pay are in millions of pesos per capita.

6.2 Insurer Marginal Costs

Table 4 presents the results for the insurer marginal cost function. We estimate it using pre-termination data to later test the supply model's out-of-sample fit and cluster standard errors at the insurer–market level. Column (1) excludes consumer type fixed effects and column (2) includes them. Both specifications include an indicator for whether the market has no hospitals, since hospital network breadth is mechanically zero in those markets. Comparing the two columns reveals the direction of the bias from unobserved selection: the marginal effects of hospital and non-hospital network breadth are larger in column (1) than column (2).

Focusing on column (2), insurers' marginal costs are increasing in hospital network breadth but decreasing in non-hospital network breadth. Insurers also enjoy economies of scope as evidenced by the negative coefficient on the interaction between the two network types. On average, a one percentage point increase in hospital network breadth raises insurers' cost per enrollee by 1,800 pesos ($0.61 of 2016). Appendix Figure 5 presents the in-sample fit of the marginal cost model and Appendix Figure 6 provides further evidence of adverse selection, showing that the marginal cost of expanding hospital network breadth is positively correlated with consumers' willingness to pay for it, while the correlation is considerably weaker for non-hospital network breadth.

TABLE 4: Insurer Marginal Cost Model

	Average cost per consumer type	
Variable	(1)	(2)
Hospital network breadth	0.050*	0.089***
	(0.040)	(0.041)
Non-hospital network breadth	0.099***	-0.021
	(0.029)	(0.028)
Hospital × Non-hospital network breadth	-0.050	-0.151***
	(0.061)	(0.062)
No hospitals in market	0.129***	0.146***
	(0.103)	(0.098)
Constant	0.261***	0.724***
	(0.060)	(0.059)
Hospital avg. marginal effect	0.034	0.018
Provider avg. marginal effect	0.113	-0.094
Insurer FE	Yes	Yes
Market FE	Yes	Yes
Consumer type FE	No	Yes
N	813,124	813,124
Pseudo-R^2	0.043	0.260

Note: Table presents non-linear least squares regression of average costs per consumer type. An observation is a consumer type-insurer-municipality-year. Column (1) includes market and insurer fixed effects. Column (2) additionally includes consumer type fixed effects. Estimation uses data from 2013 to 2017 from all municipalities in the country. The bottom of the table reports the average marginal effect of a 1 percentage point increase in hospital and non-hospital network breadth. Robust standard errors in parenthesis. $^{***}p < 0.001$,$^{**}p < 0.05$,$^{*}p < 0.1$.

6.3 Dropout and Transition Probabilities

We estimate dropout and diagnosis transition probabilities non-parametrically from the data and outside the model. For instance, the dropout probability is calculated as the share of contributory enrollees in year $t-1$ who switch to the subsidized system in year t, while the transition probability is the share of enrollees with disease x in year t who had disease y in $t-1$. To compute these probabilities we use the full sample of individuals in the 13 largest municipalities (independent of their enrollment spell lengths). Their summary statistics are presented in Appendix Tables 6 and 7.

6.4 Insurer Fixed Costs

Our model of insurer competition is static, hence we only require data from a single year to estimate the remaining parameters related to insurers' fixed costs. However, since we

have multiple years of data, our model is over-identified for these parameters. In other words, equation (6) represents a system of $J \times M \times K$ equations in the same number of unknowns, but we have $J \times M \times K \times T$ data points.

We choose to estimate our model of insurer competition using data from 2015 before SaludCoop is terminated, to later test the supply model's out-of-sample fit. We use the 2015 cross-section of individuals to forward-simulate marginal and total variable profits for $T = 100$ periods.[21] In every period and for every combination of sex, age, diagnosis, income group, insurer, incumbent insurer, and municipality, we compute demand and marginal costs (and their derivatives) using our estimates from Tables 2 and 4. Then, conditional on a consumer type θ, transitions across periods are governed by exogenous dropout probabilities, exogenous transition probabilities across diagnoses, and endogenous transition probabilities across insurers.[22]

After simulating insurers' marginal and total variable profits, we estimate the fixed cost parameters using the FOCs.[23] In all markets, marginal variable profits are positive (see in Appendix Table 8), suggesting fixed costs play a role in our characterization of insurers' decisions to offer network breadth.

Results. Table 5 presents the results separately for hospital and non-hospital networks. For both network types, the coefficient on network breadth is positive, indicating that fixed costs increase with network breadth and deter insurers from expanding their networks indefinitely. The effect is considerably larger for non-hospital than for hospital networks, consistent with the descriptive evidence in Figure 3 that showed that fixed costs per capita rise more steeply with non-hospital network breadth. This finding reflects the higher administrative burden of contracting with small individual providers than with large hospitals.

[21] We choose to forward simulate the profit function for 100 periods because the discount factor at $T = 100$ is very close to zero.

[22] Dropout probabilities depend mostly on the likelihood of becoming unemployed rather on insurers' network breadth and in Appendix Table 3 we showed that transition probabilities across diagnoses are statistically unrelated.

[23] We do not use the data on insurers' total administrative costs for this estimation because the information is aggregated to the national level and only varies over years.

TABLE 5: Insurer Fixed Cost Model

Variable	Log marginal variable profits	
	Hospital	Non-hospital
	(1)	(2)
Network breadth	2.089**	5.599***
	(0.807)	(1.235)
Constant	10.844***	7.719***
	(0.841)	(1.084)
Insurer FE	Yes	Yes
Market FE	Yes	Yes
N	77	84
R^2	0.820	0.833

Note: Table presents OLS regression of log marginal variable profits on hospital network breadth and non-hospital network breadth in columns (1) and (2), respectively. An observation in these regressions is an insurer-market. Specifications include insurer and market fixed effects. The number of observations in column (1) is smaller than in column (2) because there are 7 observations for which the marginal variable profit is negative. We keep insurer-markets associated with those observations fixed in the counterfactuals. Robust standard errors in parenthesis. $^{***}p < 0.001$,$^{**}p < 0.05$,$^{*}p < 0.1$.

6.5 Out-of-sample Fit

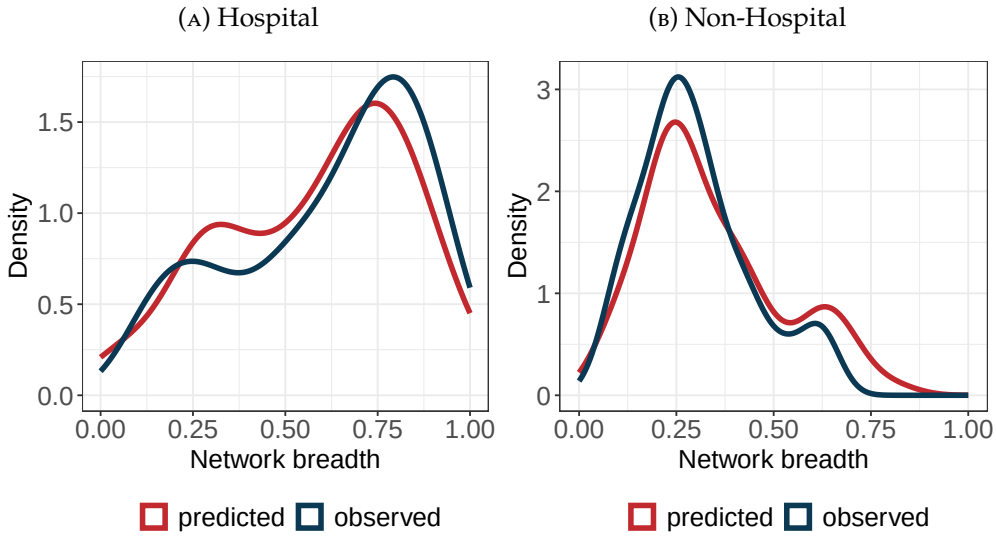
To evaluate the model's out of sample fit, we simulate insurers' network breadth choices in 2016 under the government-mandated enrollment rule that reassigned SaludCoop enrollees to Cafesalud. For these enrollees, we assume that $y_{ijm,t-1} = 1$ when assigned to Cafesalud.[24] Figure 5 compares the observed distribution of hospital and non-hospital network breadth (in dark blue) with the model's predictions (in dark red). The model closely replicates insurer's actual choices with slight over-predictions for non-hospital networks and slight under-predictions for hospital networks.

6.6 Robustness Checks

We conduct several robustness checks on our estimation. Appendix Tables 9 and 10 present a version of insurer demand and marginal costs, respectively, where we consider only the diagnoses a person received during the first year they were enrolled. These exercises address potential concerns over mechanical correlation between diagnoses and network breadth. Our estimates of network breadth slope parameters are robust to this definition of diagnoses in the demand model. In the marginal cost model, the coefficients

[24]This assumption is reasonable given that, during the 90-day grace period, the government required Cafesalud to maintain the same provider network as SaludCoop.

FIGURE 5: Out-of-Sample Prediction of Network Breadth



Note: Figure presents the distribution of observed hospital and non-hospital network breadth for 2016 (in dark blue) and the distribution of predicted hospital and non-hospital network breadth for 2016 imposing the observed reassignment rule in which SaludCoop enrollees are transferred to Cafesalud (in dark red). An observation is an insurer-market.

are slightly more sensitive though the qualitative pattern underlying our counterfactuals is preserved (marginal costs rise with hospital breadth, fall with non-hospital breadth, and exhibit economies of scope between the two).

7 Counterfactual Automatic Enrollment Rules

7.1 Setup

We compare alternative enrollment rules after SaludCoop's termination along the dimensions of network breadth, short-run average consumer surplus per capita, short-run average healthcare spending per capita, and degree of adverse selection. Short-run average consumer surplus per capita is defined as the inclusive value from the logit demand system:[25]

$$CS = \left(\sum_{ijm} s_{ijm} \right)^{-1} \left(\sum_{ijm} s_{ijm} \log(\sum_{j \in \mathcal{J}_{mt}} \exp(d_{ijmt})) \right)$$

[25]The calculation of consumer's expected utility that generates this inclusive value would have an integration constant. In our results, we interpret only changes in consumer surplus relative to the observed scenario to average-out this integration constant.

Importantly, in computing consumer surplus we count only the preference component of inertia, $\hat{\Delta}_\theta = -\hat{\phi}$, and exclude the switching cost $\hat{\kappa}_\theta = \hat{\lambda}_\theta + \hat{\phi}$, following the decomposition in Section 5.

Short-run healthcare spending per capita is given by:

$$AC = \Big(\sum_{ijm} s_{ijm} \Big)^{-1} \Big(\sum_{\theta(i)jm} C_{\theta jm} s_{ijm} \Big)$$

Finally, in the style of Einav and Finkelstein (2011), we measure the degree of adverse selection by the correlation between the consumer's willingness-to-pay for network breadth and insurers' marginal cost of increasing network breadth (as in Appendix Figure 6).

We evaluate the following enrollment rules in each market:

1. *Random*: SaludCoop enrollees are randomly assigned in equal proportions to incumbent insurers in their municipality of residence.

2. *Overlap*: SaludCoop enrollees are assigned to the incumbent insurer with the greatest hospital and non-hospital network overlap with SaludCoop in their municipality of residence.

3. *Proportional*: SaludCoop enrollees are randomly assigned to incumbent insurers in proportion to their 2015 market shares in their municipality of residence. For example, suppose Insurer A covers 30 enrollees, Insurer B covers 20 enrollees, and SaludCoop covers 50 enrollees in a market. Then, after SaludCoop's termination, Insurer A receives 30 SaludCoop enrollees ($= 50 \times \frac{30}{20+30}$) and Insurer B receives 20 ($= 50 \times \frac{20}{30+20}$).

4. *Broadest*: SaludCoop enrollees are assigned to the incumbent insurer with the broadest hospital network in their municipality of residence.

5. *Largest*: SaludCoop enrollees are assigned to the incumbent insurer with the largest market share in 2015 in their municipality of residence (excluding SaludCoop).

6. *Own choice*: Enrollees are free to choose their insurer after excluding SaludCoop from the choice set.

In these enrollment rules, we impose a default insurer only during the first year after SaludCoop's termination but allow consumers to switch after this initial period. In each rule, using as starting value the observed vector of network breadths in 2015, we compute the FOCs for each insurer and market. From these FOCs, we solve for network breadth as $H_{jm}^k = (\log(MP_{jm}^k) - \hat{v}_{jm}^k)/\tilde{\omega}^k$, which we then use as starting point in the next iteration. We iterate until the maximum residual network breadth by absolute value is less than 10^{-4}. In counterfactuals, the solution to this set of equations is likely unique given the rich preference and cost heterogeneity across insurers, but a direct proof of uniqueness is challenging in this context.

7.2 Predictions

Switching. To predict how each enrollment rule will affect equilibrium network breadth, we begin by characterizing whether healthy or sick consumers will be more likely to switch in each scenario. Specifically, we compute, among SaludCoop enrollees in the first year following reassignment, the covariance between a measure of switching propensity and a measure of per capita profit that captures heterogeneity by health status but abstracts from network breadth. A positive covariance would indicate that the rule induces greater switching among higher-profit individuals. Hence, insurers will expand their networks to attract them. Conversely, a negative covariance would imply that lower-profit individuals are more likely to switch, and insurers will reduce network breadth to avoid attracting them.

We compute a proxy of switching propensity as the difference between the consumer's match value with SaludCoop and their assigned insurer, divided by the inertia parameter:

$$p_i = \frac{\xi_{\theta(i),\text{SaludCoop}} - \xi_{\theta(i),\text{assigned}}}{\lambda_{\theta(i)}}$$

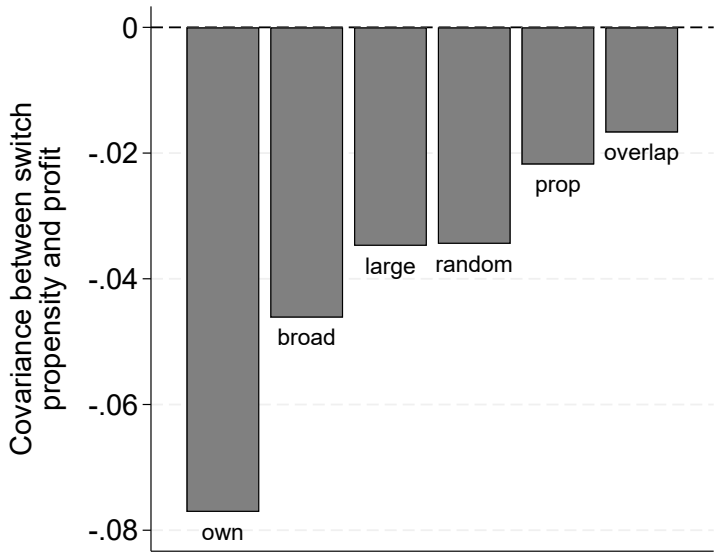
Intuitively, if the match value with SaludCoop is higher than the match value with the assigned insurer, the individual will be more likely to switch; and this effect is amplified if the consumer has relative low value of inertia.

We calculate the per capita profit abstracting from provider networks, as:

$$\pi_i = R_{\theta(i)m} - C_{\theta(i)m,\text{assigned}} = R_{\theta(i)m} - e^{\gamma_\theta + \eta_m + \delta_j}$$

Figure 6 shows a clear ranking of the enrollment rules in terms of the covariance between p_i and π_i. All rules generate a negative covariance, indicating that consumers who are more likely to switch tend to be less profitable. Assignment based on network overlap and the two rules with a random assignment component (random and proportional assignment) – where mismatches between consumers and their assigned insurer are most likely – produce the least negative covariances, meaning they disproportionately induce switching among relatively more profitable consumers. Thus, rules with random components and network overlap will be more likely to increase network breadth in equilibrium, while own choice rules will tend to reduce it.

FIGURE 6: Covariance Switching Metric



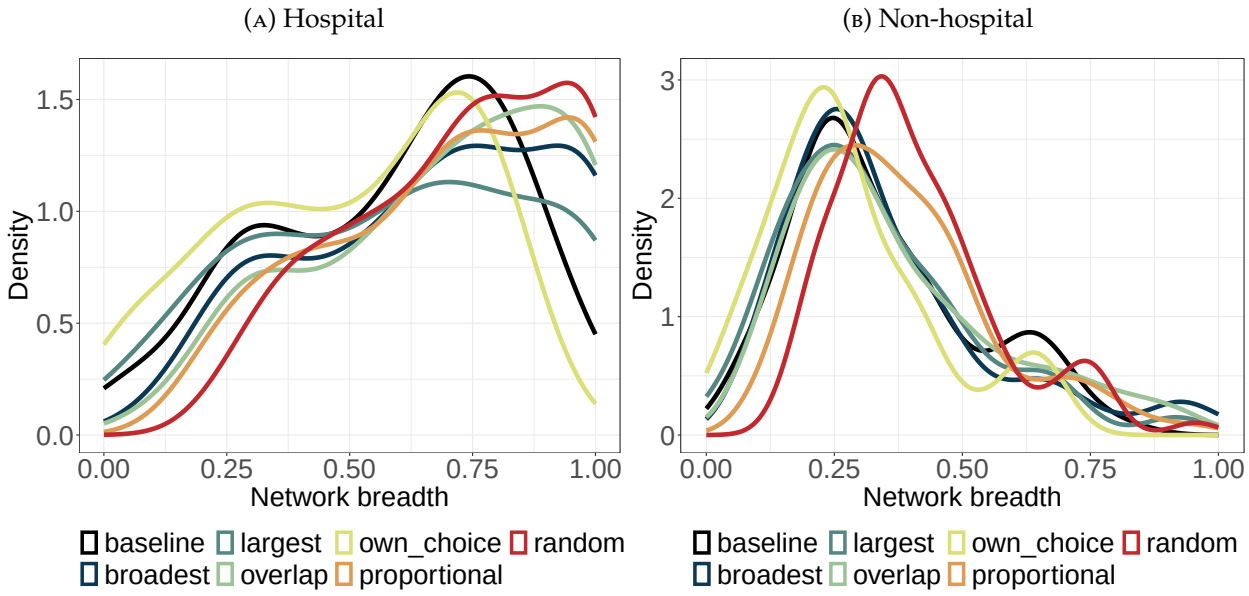
Note: Figure shows the covariance between the switching propensity and the marginal cost in absence of network breadth among SaludCoop enrollees in the first year after the reassignment. Switching propensity is calculated as the difference between the idiosyncratic match value with SaludCoop $\xi_{\theta(i),\text{SaludCoop}}$ and the assigned insurer $\xi_{\theta(i),\text{assigned}}$, divided by the consumer's value of inertia $\lambda_{\theta(i)}$. The marginal cost, abstracting from network breadth, is the prediction from equation (5) assuming $H_{jm} = 0$.

Still, this exercise abstracts from equilibrium network breadth, so we now turn to the full simulation.

7.3 Results

Figure 7 presents the counterfactual distribution of hospital network breadth in Panel A and non-hospital network breadth in Panel B under each enrollment rule, and Table 6 reports the averages weighting by demand. Four rules – random, proportional, overlap, and broadest – raise average hospital network breadth by 23–24% relative to baseline, and non-hospital breadth by 9–12%. In contrast, the own-choice rule lowers average network breadth by 8% and 15% for each type respectively. Allowing consumers to choose freely without a default minimizes the pool of profitable consumers willing to switch, which weakens competition for switchers and leads insurers to narrow their networks. Free consumer choice is therefore not necessarily welfare-enhancing in this setting once insurers' responses are taken into account.

FIGURE 7: Counterfactual Network Breadth



Note: Figure presents the distribution of hospital and non-hospital network breadth under each enrollment rule. We depict the model's prediction of the observed assignment rule in black, labelled "baseline."

The rules differ more in the distribution of network breadth than in the mean. Random enrollment is the only rule that eliminates narrow hospital networks: under the baseline scenario and every other rule, a substantial share of insurer-markets have hospital breadth below 0.5, whereas under random enrollment that mass falls substantially and the distri-

bution concentrates between 0.75 and 1. For non-hospital networks, random enrollment shifts the entire distribution to the right, with its mode at 0.35 against 0.25 under the baseline scenario, while proportional, overlap, and broadest raise the mean mainly by thickening the upper tail. Own choice generates the narrowest networks of both types across the entire distribution.

TABLE 6: Additional Outcomes Under Counterfactual Enrollment Rules

Scenario	Hosp. network	Non-hosp. network	Marginal cost*	Hosp. adverse selection	Non-hosp adverse selection
	(1)	(2)	(3)	(4)	(5)
baseline	0.624	0.375	1.104	0.356	0.407
broadest	0.769	0.413	1.395	-0.082	0.388
largest	0.698	0.385	1.410	-0.076	0.405
overlap	0.772	0.420	1.389	-0.086	0.389
own choice	0.572	0.318	1.407	-0.064	0.378
proportional	0.772	0.409	1.368	-0.083	0.371
random	0.768	0.412	1.355	-0.092	0.376

Note: Table presents additional counterfactual outcomes for each enrollment rule. Columns (1) and (2) present the demand-weighted hospital and non-hospital network breadth across insurers. Column (3) presents the average marginal cost across insurers, which can be interpreted as the healthcare spending per capita. Column (4) presents the correlation between willingness-to-pay for hospital networks and marginal costs. Column (5) presents the same measure as (4) but for non-hospital networks. (*) measured in millions of COP. The average exchange rate in 2016 was 3,050 COP/USD.

Table 6 also shows that every rule raises marginal cost per enrollee relative to baseline, but random enrollment raises it the least despite producing the largest shift in both network distributions. Random enrollment generates the largest reduction in the correlation between marginal cost and willingness-to-pay for hospital and non-hospital networks, which is why its cost increase is relatively small.

Table 7 summarizes other outcomes of interest. Column (1) reports consumer surplus without allowing insurers to respond endogenously to the rules, while column (2) reports consumer surplus allowing for endogenous network breadth supply responses. Column (3) reports insurers' variable profit per enrollee and column (4) reports social surplus (consumer plus insurer surplus), all allowing for endogenous supply responses.

Holding the supply side fixed, the rules are nearly indistinguishable: consumer surplus ranges from 1.69 to 1.71 million pesos ($572–$579) across all six, with random enrollment last and the largest and own-choice rules first. Allowing insurers to respond reverses this ordering and widens the gaps. Random, proportional, and overlap enrollment now

generate the highest consumer surplus, 1.94–1.95 million pesos or 47% above baseline. The largest and own-choice rules fall to the bottom, with consumer surplus at 1.79 and 1.59 million pesos ($606 and $538). Thus, failing to account for supply-side responses leads to fundamentally different conclusions about which enrollment rules improve consumer welfare. Finall, random enrollment also generates the highest social surplus, 2.36 million pesos or 44% above baseline. The own-choice rule transfers surplus from consumers to insurers and generates the lowest social surplus.

TABLE 7: Surplus Under Counterfactual Enrollment Rules

Scenario	Consumer surplus without supply (1)	Consumer surplus with supply (2)	Variable profit per enrollee (3)	Social surplus (4)
baseline	—	1.319	0.326	1.644
broadest	1.698	1.927	0.409	2.336
largest	1.712	1.785	0.404	2.189
overlap	1.698	1.946	0.411	2.357
own choice	1.708	1.586	0.433	2.018
proportional	1.691	1.949	0.413	2.362
random	1.685	1.944	0.419	2.364

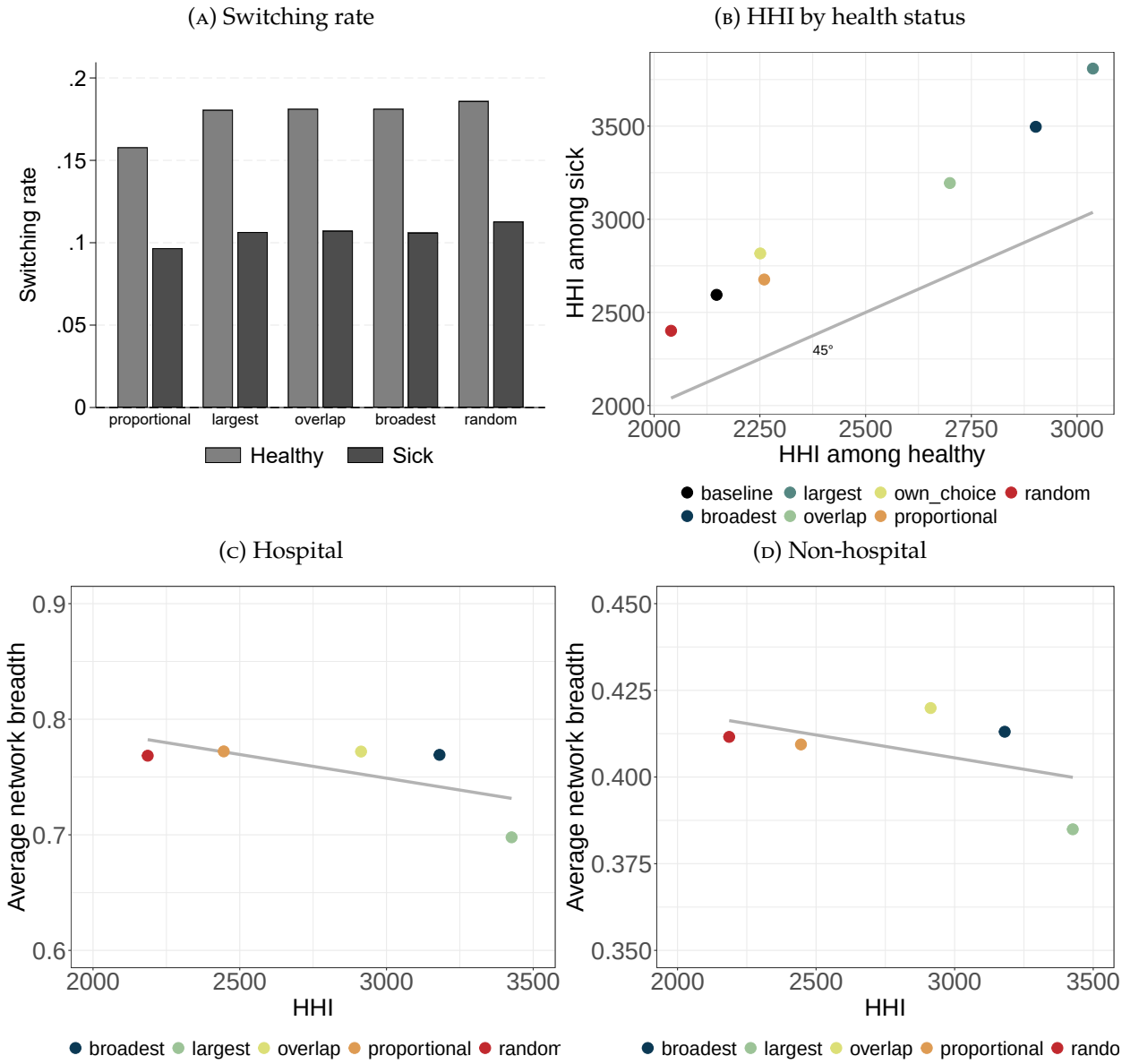
Note: Table presents additional counterfactual outcomes for each enrollment rule. Column (1) shows the demand-weighted average consumer surplus per capita without allowing for endogenous supply responses. Column (2) shows the same measure of consumer surplus allowing the supply-side to respond endogenously. Column (3) presents the average variable profit per enrollee. Column (4) presents social surplus, the sum of consumer and insurer surplus. All colums are measured in millions of COP. The average exchange rate in 2016 was 3,050 COP/USD.

7.4 Analysis of Mechanisms

As shown before, our model offers two explanations for why random enrollment outperforms other rules in terms of social surplus and other outcomes: inertia and adverse selection. In terms of inertia, the random assignment rule maximizes the set of mismatched consumers, and hence the set of those who are likely to switch. Table 3 showed that the value of inertia for healthy consumers is smaller than for sick consumers. Therefore, incumbent insurers will expand their networks to attract the healthy. Although sick consumers value hospital network breadth almost as much as healthy ones, in relative terms, sick consumers value a 10 percentage point increase in hospital network breadth less than what they value remaining with their incumbent insurer (see columns 1 and 2 of Table 3). This suggests that by expanding their networks insurers will attract more

healthy consumers than sick ones.

FIGURE 8: The Role of Consumer Inertia and Selection in Insurer Competition



Note: Panel A presents the switching rate among the healthy and the sick for each enrollment rule among SaludCoop enrollees. We exclude the observed and own-choice rules because switching rates in those scenarios equal 100%. Panel B presents a scatter plot of average HHI conditional on individuals without diseases ("healthy") against average HHI conditional on individuals with chronic diseases ("sick"). The gray line is the 45 degree line. Panel C presents a scatter plot of average HHI across markets against average hospital network breadth across insurers and markets for each enrollment rule and Panel D against average non-hospital network breadth. The dashed gray line corresponds to a linear fit.

Figure 8, Panel A confirms these predictions: switching rates out of the assigned insurer are higher under random enrollment than under all other rules, particularly among healthy

consumers.[26] Importantly, healthy consumers switch at higher rates than sick consumers under every rule, but the gap is largest under random enrollment.

In terms of adverse selection, random enrollment spreads the health risk more evenly across insurers because sick consumers are more likely to stick with their assigned health plan. Therefore, we can expect insurer market concentration among sick patients to be lower than under other enrollment rules. Indeed, Figure 8, Panel B shows that only random enrollment reduces market concentration among individuals with and without diagnoses relative to the baseline scenario.

Panels C and D connect these concentration patterns to network breadth outcomes. Automatic enrollment rules that generate lower HHI are associated with broader hospital and non-hospital networks, while rules that increase market concentration are associated with narrower networks.

Who switches has important implications for equilibrium outcomes. Under the own-choice rule, all SaludCoop consumers shop for a new insurer in the first year after the termination since there is no incumbent relationship to generate inertia, and network breadth disproportionately attracts sick consumers who value hospital access most, hence insurers reduce network breadth. Indeed, in prior work (Buitrago et al., 2026), we showed that when sick consumers are the marginal switchers, insurers drop high-cost providers specializing in chronic care (such as public hospitals and cancer centers), possibly to minimize costs and deter further enrollment of unprofitable patients. This explains why the own-choice rule generates the narrowest networks in equilibrium despite giving consumers the most freedom to choose.

7.5 Discussion

Scale of the reassignment. A natural question is whether the insurer responses that shift the ranking of the enrollment rules depend on the reassigned pool being large (former SaludCoop enrollees were nearly a fifth of the market). Although the size of the

[26] We omit the own-choice and observed rules since switching rates are mechanically equal to one in both cases. In the own-choice rule we consider the "assigned" insurer to be SaludCoop, hence naturally everyone switches out. In the observed rule, among the continuously enrolled, all SaludCoop enrollees end up leaving Cafesalud.

pool plausibly scales the magnitude of the network response, the broader implication of our results is that the impact of automatic enrollment rules on welfare depends on the distribution of inertia and preferences for plan characteristics across consumers, not just on the quality of the initial match.

A large pool of reassigned individuals is also not unusual. For example, in Medicare Advantage in the U.S., plan terminations are frequent (Abaluck et al., 2021) and, because plans compete at the county level, a terminated plan can account for a large share of a county's enrollees, as SaludCoop did in many municipalities. In Minnesota, the 2025 shutdown of a single insurer displaced more than 150,000 beneficiaries, and more than 60% of the state's Medicare Advantage enrollees must change plans for 2027 because their plan is terminating or leaving their county—a striking figure compared to the national average of 13%.[27] The question our results speak to is how enrollment for these displaced enrollees should be handled.

Automatic enrollment as a nudge. Equilibrium outcomes under the counterfactual enrollment policies we consider arise from imposing a default insurer only during the first year after SaludCoop's termination. The fact that a one-year policy has lasting effects on the distribution of network breadth and consumer surplus implies that the dynamic consequences of inertia can be reversed with short-run policies. Random enrollment essentially nudges consumers into active plan choice resulting in improved outcomes once supply-side decisions are taken into account. These results complement the literature on the welfare effects of behavioral nudges (e.g., Domurat et al., 2021; Handel, 2013).

8 Conclusions

This paper shows that automatic enrollment in health insurance impacts insurer competition and long-run equilibrium outcomes by determining which consumers are more likely to switch after reassignment. To show this we use data from Colombia's contributory

[27] See *The Minnesota Star Tribune*, "Why Minnesota seniors face the nation's biggest health insurance shake-up," October 2026, https://www.startribune.com/why-minnesota-seniors-face-the-nations-biggest-health-insurance-shake-up/601901053.

healthcare system where insurers compete mainly on their network of covered providers. The largest health insurer in this health system, which covered 20% of enrollees, was terminated by the government in December 2015. The termination provides the variation we need to identify the components of inertia and of preferences for hospital and non-hospital networks which determine switching.

To compare counterfactual enrollment rules after the termination, we develop and estimate an equilibrium model of insurer competition on hospital and non-hospital network breadth. We find that enrolling patients randomly to incumbent insurers in equal proportions is effective at increasing social surplus and both types of network breadth relative to other rules such as enrollment to the largest insurer or to the insurer with the broadest networks. We show that the main reason why random enrollment outperforms other rules is that it disproportionately encourages switching among healthy consumers. This incentivizes insurers to compete more aggressively for these profitable patients, rather than resorting to harvesting their stock of enrollees. Importantly, we show that failing to endogenize insurers' network breadth responses to the automatic enrollment rules leads to different conclusions about the rules that improve social welfare.

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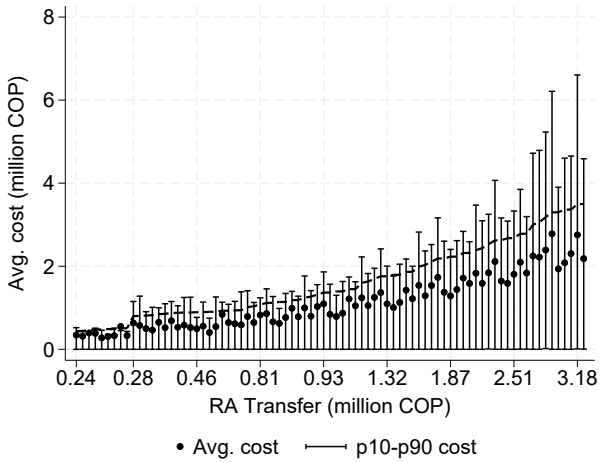
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Supplemental Appendix

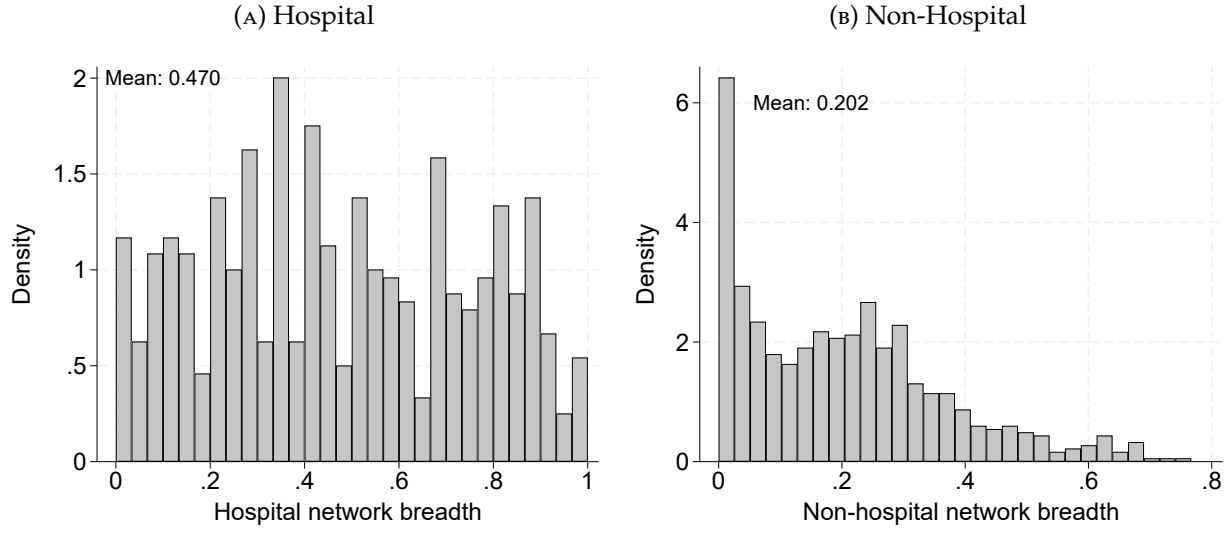
Appendix A Additional Descriptives and Results

Appendix Figure 1: Risk-Adjustment Fit Among Continuously Enrolled



Note: Figure presents the mean (black circles), and the 10th and 90th percentiles (error bars) of healthcare cost per risk-adjusted transfer among the continuously enrolled. The dashed line is the 45 degree line.

Appendix Figure 2: Distribution of Municipal Network Breadth



Note: Figure shows the distribution of hospital network breadth in Panel A and non-hospital network breadth in Panel B. An observation in these histograms is an insurer-market-year conditional on the 13 largest municipalities and for insurers that operate in the contributory system. In Panel A, we condition to markets that have at least one hospital. Hospitals are identified as institutions with an intensive care unit.

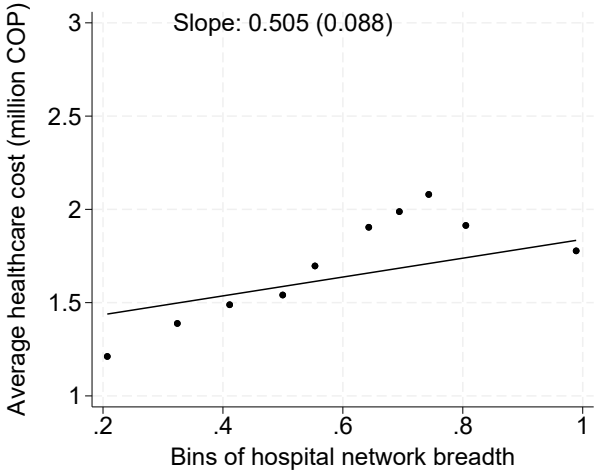
Appendix Table 1: Summary Statistics of Main Samples

Variable	(1) Full		(2) Continuous		(3) 13 Municipalities	
	mean	sd	mean	sd	mean	sd
Male	0.476	(0.499)	0.471	(0.499)	0.516	(0.500)
Age	43.26	(16.73)	44.01	(16.93)	44.01	(16.40)
Earnings†	0.804	(1.296)	0.816	(1.336)	1.225	(1.672)
Cancer	0.083	(0.276)	0.084	(0.278)	0.071	(0.257)
Cardiovascular disease	0.179	(0.383)	0.186	(0.389)	0.127	(0.333)
Diabetes	0.030	(0.170)	0.031	(0.174)	0.021	(0.143)
Pulmonary disease	0.013	(0.113)	0.014	(0.116)	0.010	(0.099)
Renal disease	0.009	(0.097)	0.010	(0.099)	0.008	(0.087)
Other disease	0.076	(0.265)	0.077	(0.267)	0.052	(0.221)
Healthy (No diagnoses)	0.565	(0.496)	0.552	(0.497)	0.573	(0.495)
Total healthcare cost†	0.786	(4.379)	0.822	(4.516)	0.792	(1.188)
OOP cost†	0.145	(0.420)	0.150	(0.434)	0.157	(0.126)
Individuals x Years	75,918,492		68,328,039		28,698,816	

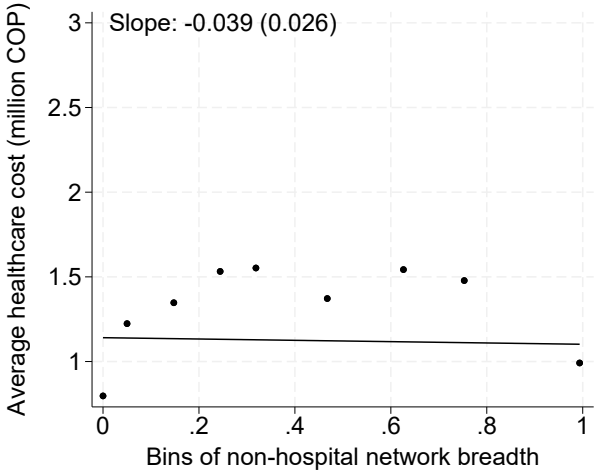
Note: Table presents the mean and standard deviation in parenthesis of each variable. Column (1) uses the full sample of individuals aged 19 or older who were enrolled with an insurer in the contributory system between 2013 and 2017. Column (2) uses the subsample of continuously enrolled individuals defined as those without gaps in enrollment between the first and last years we observe them in the enrollment data. Column (3) uses the subsample of continuously enrolled individuals who reside in one of the 13 largest municipalities: Bogotá, Medellín, Cali, Barranquilla, Bucaramanga, Manizales, Pereira, Cúcuta, Pasto, Ibagué, Montería, Cartagena, and Villavicencio. (†) measured in millions of COP of 2014.

Appendix Figure 3: Correlation Between Average Cost, Fixed Cost Per Capita, and Network Breadth

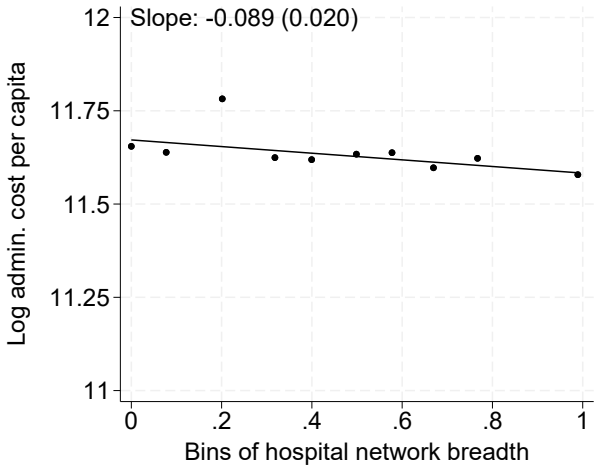
(a) Average Cost: Hospital



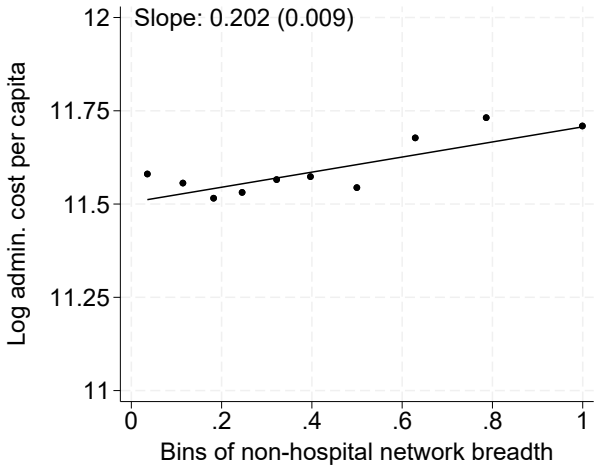
(b) Average Cost: Non-hospital



(c) Fixed Cost per Capita: Hospital

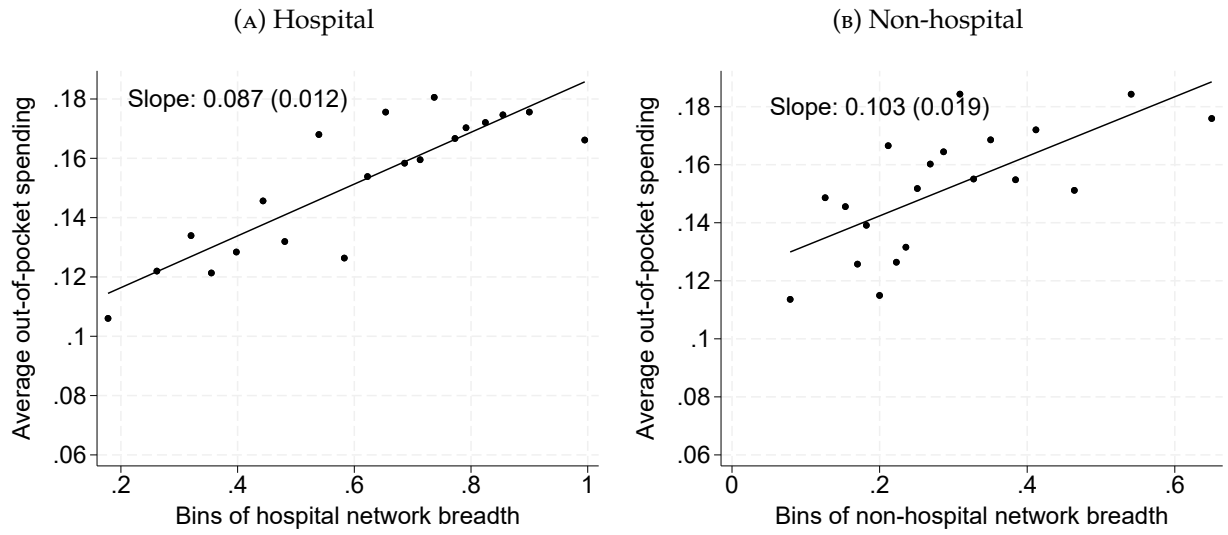


(d) Fixed Cost per Capita: Non-hospital



Note: Panels A and B present a scatter plot of the average healthcare cost by 20 equally sized bins of hospital network breadth and non-hospital network breadth, respectively. Panels C and D present a scatter plot of the log administrative cost per capita by 20 equally spaced bins of hospital network breadth and non-hospital network breadth, respectively. To construct Panels A and B, we aggregate our preferred sample to the consumer type-insurer-market-year level, calculating average healthcare costs and hospital and non-hospital network breadths. To construct Panels C and D, we use insurer-year-level income statements reporting administrative costs and divide this by their total number of enrollees. Solid lines represent linear fits and the numbers in the upper corner of each panel report the slope coefficient and standard error from this regression.

Appendix Figure 4: Correlation Between Out-of-Pocket Costs and Network Breadth



Note: Figure presents a scatter plot of the average hospital and non-hospital network breadth by 20 equally sized bins of the average Charlson index. To construct this figure, we aggregate our preferred sample to the insurer-market-year level, calculating average Charlson index and hospital and non-hospital network breadth. Then, we calculate 20 equally sized bins of the Charlson index. Finally, we compute the average of each type of network breadth per bin. Solid lines represent linear fits.

Appendix Table 2: Placebo Test on Network Breadth Commitment

	$\Delta H^{\text{hosp}}_{jmt}$ (1)	$\Delta H^{\text{non-hosp}}_{jmt}$ (2)
$\%\Delta N_{jm,t+1}$	-0.000817	0.00484
	(0.00190)	(0.00259)
$H^{\text{hosp}}_{jm,t-1}$	-0.176***	
	(0.0516)	
$H^{\text{non-hosp}}_{jm,t-1}$		-0.208***
		(0.0264)
Constant	0.0207**	0.138***
	(0.00762)	(0.0144)
Market FE	Yes	Yes
Insurer FE	Yes	Yes
N	1607	1607
R^2	0.675	0.552

Note: Table reports OLS regressions using as outcome variables the change in hospital and non-hospital network breadth over time. The main independent variable is the lead change in enrollment. An observation is an insurer-market-year. Estimations use data from 2015 before SaludCoop's termination to test whether insurers who gained enrollees after the termination where not on different network breadth trends before the termination. All specifications include insurer and market fixed effects. Standard errors are clustered at the market level. $^{***}p < 0.001, ^{**}p < 0.05, ^{*}p < 0.1$.

Appendix Table 3: Conditional Independence of Transition Probabilities and Network Breadth

	Any diagnosis
Hospital network breadth	0.133
	(0.153)
Non-hospital network breadth	-0.457
	(0.332)
Constant	0.241***
	(0.0752)
Insurer-Market FE	Yes
Year FE	Yes
Observations	1,084,259
R^2	0.0398

Note: Table presents OLS regression of an indicator for having any chronic disease on hospital and non-hospital network breadth. Specification includes insurer-municipality and year fixed effects. Estimation uses the random sample for model estimation conditional on observed choices and consumers who did not have a chronic disease in $t-1$. Standard errors in parenthesis are clustered at the insurer level. $^{***}p < 0.001, ^{**}p < 0.05, ^{*}p < 0.1$.

Appendix Table 4: First-Stage Regressions of Demand

	OOP spending (1)	Hospital network breadth (2)
Coins rate * Provider exit rate from network	11.200***	2.989
	(3.291)	(3.572)
Hospital exit rate from network	-0.318***	-0.147
	(0.141)	(0.195)
Any public provider in network	0.011**	0.108***
	(0.007)	(0.040)
Insurer-male FE	Yes	Yes
Insurer-diagnosis FE	Yes	Yes
Insurer-age group FE	Yes	Yes
Insurer-market FE	Yes	Yes
Observations	13,878,827	13,878,827
R^2	0.208	0.944

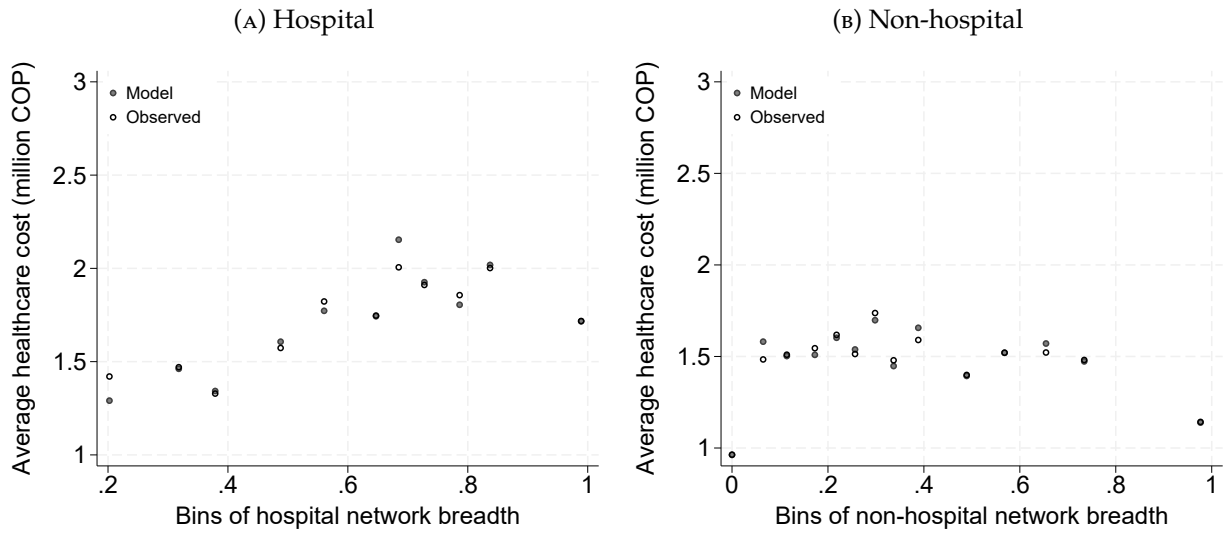
Note: Table presents the control function for out-of-pocket spending in column (1), for hospital network breadth in column (2), and for non-hospital network breadth in column (3). To estimate each regression, we collapse the insurer choice data to the level at which each variable varies: consumer type-insurer-municipality-year for out-of-pocket spending and insurer-municipality-year for network breadth. Hausman instruments (out-of-pocket spending, hospital and non-hospital network breadth in other markets) are computed using data for all the country without restricting to the 13 largest municipalities. Robust standard errors in parenthesis.

Appendix Table 5: Insurer Market Shares

	Observed	Predicted
Insurer A	1.725	1.941
Insurer B	11.178	10.138
Insurer C	8.502	8.327
Insurer D	9.728	9.529
Insurer E	9.024	9.522
Insurer G	12.343	11.428
Insurer H	1.689	1.616
Insurer I	6.647	6.210
Insurer J	9.838	9.928
Insurer K	8.763	9.008
Insurer L	3.424	3.465
Insurer M	3.502	2.867
Insurer N	13.635	16.022

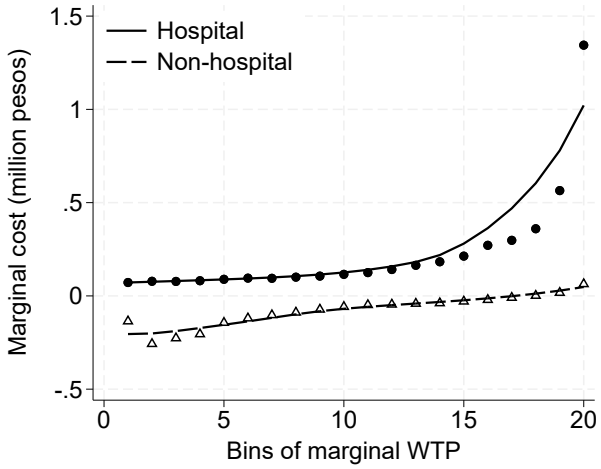
Note: Table presents observed and model predicted insurer market shares in the 13 largest municipalities using estimates from the insurer demand model. Consumers' discrete choice is simulated by drawing type-I extreme value shocks. We set the control function coefficients to zero to make these predictions.

Appendix Figure 5: Marginal Cost Model In-Sample Fit



Note: Figure presents a scatter plot of observed and predicted average cost per enrollee by 20 equally sized bins of hospital network breadth in panels A and B, and non-hospital network breadth in panels C and D.

Appendix Figure 6: Model Evidence of Adverse Selection



Note: Figure shows the insurer's short-run marginal cost in thousand of pesos $\partial C_{\theta jmt}(\mathbf{H}_{jmt})s_{ijmt}(\mathbf{H}_{mt})/\partial H^k_{jmt}$ averaged within 2-equally sized bins of consumer marginal willingness-to-pay for an additional percent in hospital and non-hospital network breadth calculated as $-\alpha_\theta^{-1}(\partial s_{ijmt}/\partial H^k_{jmt})$.

Appendix Table 6: Annual Transition Probabilities Across Diagnoses

		Cancer	Cardio	Diabetes	Pulmonary	Renal	Other	Healthy
Cancer	Mean	0.508	0.147	0.015	0.027	0.017	0.049	0.236
	SD	0.172	0.152	0.016	0.082	0.072	0.042	0.179
Diabetes	Mean	0.041	0.670	0.025	0.021	0.021	0.056	0.165
	SD	0.096	0.237	0.067	0.036	0.047	0.133	0.198
Cardio	Mean	0.035	0.165	0.613	0.018	0.022	0.032	0.115
	SD	0.076	0.104	0.166	0.031	0.030	0.043	0.133
Pulmonary	Mean	0.048	0.162	0.016	0.494	0.011	0.069	0.200
	SD	0.044	0.090	0.010	0.169	0.021	0.076	0.168
Renal	Mean	0.049	0.242	0.034	0.017	0.468	0.049	0.142
	SD	0.104	0.184	0.042	0.028	0.137	0.035	0.152
Other	Mean	0.050	0.163	0.015	0.028	0.017	0.492	0.235
	SD	0.056	0.141	0.014	0.038	0.078	0.179	0.180
Healthy	Mean	0.039	0.085	0.010	0.011	0.003	0.032	0.820
	SD	0.094	0.123	0.062	0.017	0.004	0.071	0.175

Note: Table presents mean and standard deviation in parenthesis of non-parametric estimates of annual transition probabilities across diagnoses. Uses data from 2013 to 2017.

Appendix Table 7: Summary Statistics of Annual Dropout Probabilities

	Mean	SD
Female	0.079	0.037
Male	0.091	0.040
Age 19-24	0.172	0.036
Age 25-29	0.130	0.020
Age 30-34	0.102	0.015
Age 35-39	0.090	0.015
Age 40-44	0.083	0.014
Age 45-49	0.076	0.014
Age 50-54	0.070	0.015
Age 55-59	0.063	0.016
Age 60-64	0.054	0.017
Age 65-69	0.049	0.016
Age 70-74	0.054	0.019
Age 75 or more	0.075	0.030
Cancer	0.090	0.032
Diabetes	0.076	0.038
Cardiovascular	0.073	0.036
Pulmonary	0.089	0.034
Renal	0.078	0.026
Other disease	0.074	0.037
Healthy	0.113	0.054

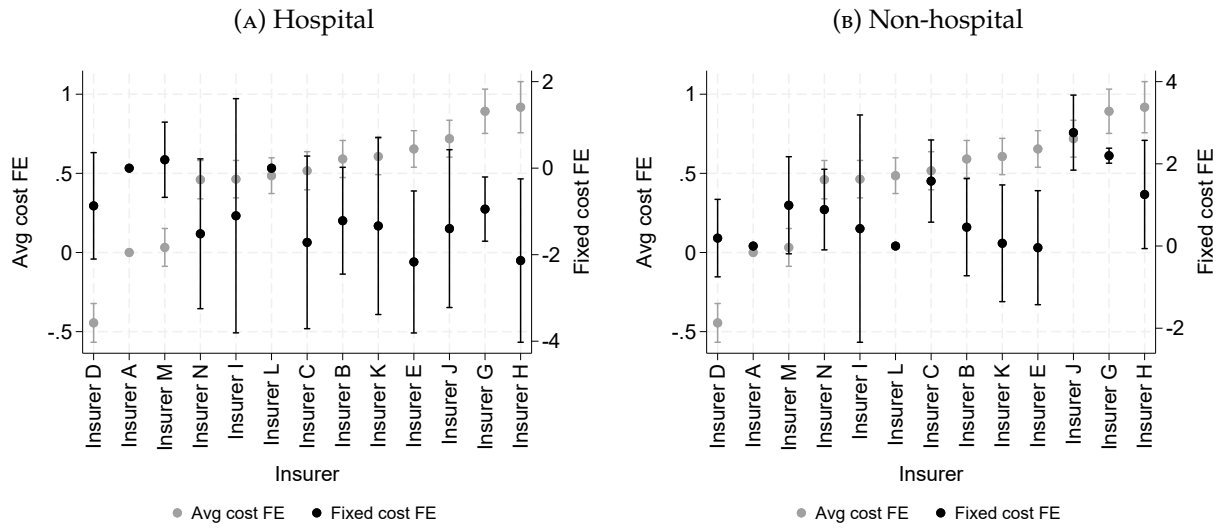
Note: Table presents mean and standard deviation in parenthesis of non-parametric estimates of the annual probability of dropping out of the contributory system. Uses data from 2013 to 2017.

Appendix Table 8: Insurer Marginal Variable Profits

Insurer	Mean	SD
Insurer A	122575.5	127800.6
Insurer B	42725.04	72561.26
Insurer C	27075.48	74473.85
Insurer D	19846.37	46454.4
Insurer E	282538.1	176739.5
Insurer G	47172.47	83082.43
Insurer H	45929.85	40365.06
Insurer J	38787.63	46793.29
Insurer K	66825.16	151417.6
Insurer L	27288.04	33036.16
Insurer M	121712.5	169698.1
Insurer N	67452.76	128517.9

Note: Table presents mean and standard deviation of marginal variable profits. Measured in millions of pesos.

Appendix Figure 7: Insurer Fixed Effects in Marginal and Administrative Costs



Note: Figure shows the estimates insurer fixed effects from the marginal cost model in gray (left vertical axis) and from the fixed cost model in black (right vertical axis).

Appendix B Robustness Checks

Appendix Table 9: Insurer Demand Model with First-Year Diagnoses

	(1) Main		(2) First diagnosis	
Variable	coef	se	coef	se
Hospital network breadth	7.942	(0.216)	8.735	(0.188)
Non-hospital network breadth	0.838	(0.102)	0.804	(0.099)
OOP spending	-5.531	(0.391)	-4.584	(0.398)
Incumbent	4.076	(0.023)	4.144	(0.021)
Incumbent SC	-0.230	(0.037)	-0.255	(0.036)
Hospital network breadth				
× Male	-0.089	(0.051)	-0.210	(0.052)
× Age	-0.006	(0.002)	0.003	(0.002)
× Cancer	0.465	(0.089)	-0.211	(0.101)
× Diabetes	1.040	(0.241)	-0.213	(0.227)
× Cardiovascular	1.097	(0.087)	0.032	(0.090)
× Pulmonary	0.316	(0.329)	-0.659	(0.329)
× Renal	0.787	(0.376)	-0.651	(0.390)
× Other disease	0.678	(0.129)	-0.047	(0.124)
Non-hospital network breadth				
× Male	-0.335	(0.059)	-0.181	(0.063)
× Age	0.008	(0.002)	-0.000	(0.002)
× Cancer	-1.402	(0.111)	-0.175	(0.123)
× Diabetes	-0.318	(0.259)	0.581	(0.273)
× Cardiovascular	-0.848	(0.115)	0.174	(0.109)
× Pulmonary	0.363	(0.417)	0.748	(0.388)
× Renal	-0.609	(0.385)	0.016	(0.468)
× Other disease	-0.631	(0.154)	-0.089	(0.154)
OOP spending				
× Male	-0.214	(0.051)	-0.085	(0.050)
× Age	0.004	(0.002)	0.002	(0.001)
× Cancer	1.169	(0.119)	0.147	(0.115)
× Diabetes	0.929	(0.147)	0.248	(0.146)
× Cardiovascular	1.229	(0.117)	0.210	(0.118)
× Pulmonary	1.033	(0.136)	0.331	(0.137)
× Renal	0.764	(0.116)	0.226	(0.120)
× Other disease	1.172	(0.136)	0.549	(0.115)
Incumbent				
× Male	0.136	(0.012)	0.119	(0.012)
× Age	0.035	(0.001)	0.034	(0.001)
× Cancer	0.268	(0.027)	0.160	(0.024)
× Diabetes	0.727	(0.055)	0.610	(0.064)
× Cardiovascular	0.401	(0.031)	0.200	(0.026)
× Pulmonary	0.881	(0.090)	0.549	(0.083)
× Renal	1.125	(0.124)	0.742	(0.107)
× Other disease	0.529	(0.032)	0.149	(0.029)
CF OOP spending	4.634	(0.383)	4.154	(0.374)
CF Hospital network	-7.656	(0.207)	-8.531	(0.178)
N	13,878,827		13,878,827	
Pseudo-R^2	0.813		0.812	

Note: Table presents maximum likelihood estimates of the insurer demand model using a conditional logit. Column (1) presents our main model. Column (2) present a model where individuals are marked as having a diagnosis if they received one in the first year of enrollment. Estimation uses a random sample of 500,000 individuals enrolled throughout the sample period from 2013 to 2017 in the 13 main capital cities or municipalities: Bogotá, Medellín, Cali, Barranquilla, Bucaramanga, Manizales, Pereira, Cúcuta, Pasto, Ibagué, Montería, Cartagena, and Villavicencio. Specification includes insurer-by-age group, insurer-by-sex, and insurer-by-diagnosis fixed effects. Boostrap standard errors in parenthesis based on 50 resamples of individuals.

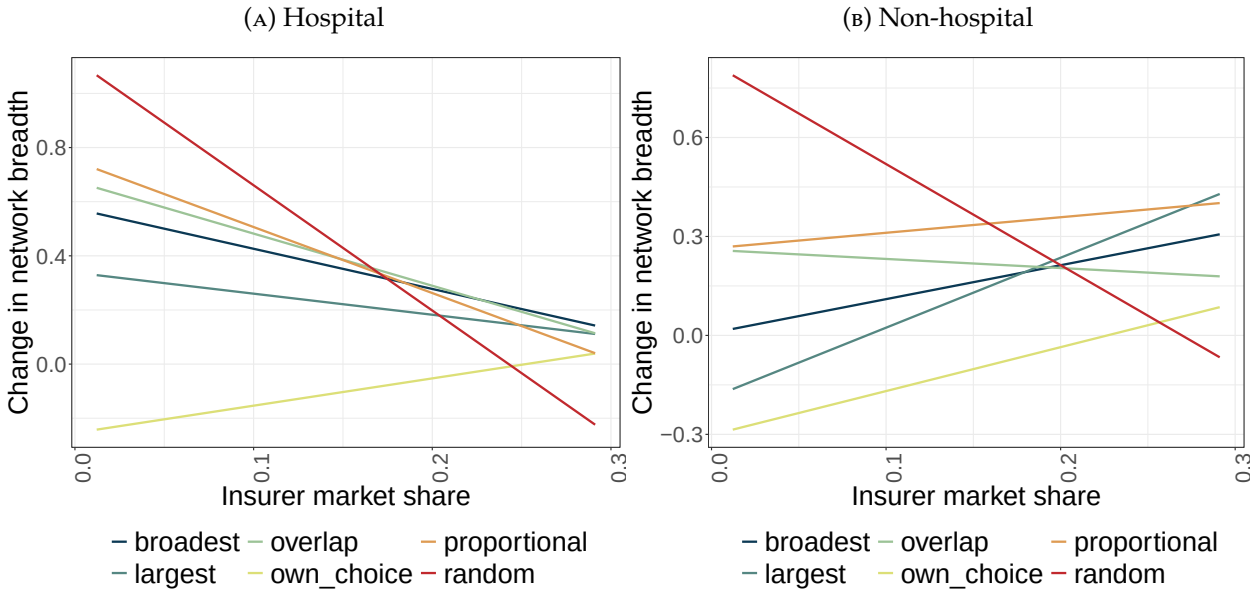
Appendix Table 10: Insurer Marginal Cost Model with First-Year Diagnosis

	(1) Main		(2) First diagnosis	
Variable	coef	se	coef	se
Hospital network breadth	0.089	(0.028)	0.046	(0.027)
Non-hospital network breadth	-0.021	(0.023)	-0.022	(0.022)
Hospital x Non-hospital network breadth	-0.151	(0.045)	-0.092	(0.044)
No hospitals in market	0.146	(0.051)	-0.017	(0.049)
Constant	0.724	(0.032)	0.905	(0.030)
Insurer FE	Yes		Yes	
Consumer type FE	Yes		Yes	
Market FE	Yes		Yes	
N	813124		757044	
Pseudo-R^2	0.260		0.269	

Note: Table presents non-linear least squares regression of average costs per consumer type. Column (1) presents the main model. Column (2) presents model in which individuals are marked as having a diagnosis if they received one in their first enrollment year. An observation is a combination of consumer type, insurer, municipality, and year. Specifications include insurer, consumer type, and municipality fixed effects. Estimation uses data from 2013 to 2015 from all municipalities in the country. Robust standard errors in parenthesis.

Appendix C Additional Counterfactual Results

Appendix Figure 8: Counterfactual Network Breadth per Insurer



Note: Figure shows a linear fit between the change in network breadth and the insurer's market share in each counterfactual enrollment rule. Panel A focuses on hospital networks and Panel B in non-hospital networks.